

QueryBook — Drawings, Reference Numerals & Brief Description of the Drawings

Scope: the complete, current QueryBook figure set — FIG. 1–FIG. 34 — as one unified set: the integral language & speech subsystem (LEL; FIG. 1–8, 31–32), the QueryBook platform (FIG. 9–30), and the Language & Identity Linking (LIL) voiceprint & identity subsystem (FIG. 33–34). **Companion** to the consolidated *QueryBook Provisional Patent Application*; the figures, reference numerals, and brief descriptions here are identical to those in that application. **Document date:** October 1, 2026. Draft — to be finalized to USPTO drawing standards (37 CFR 1.84) by a patent draftsman before filing.

Brief Description of the Drawings

FIG. 1 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 2 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 3 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 4 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 5 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 6 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 7 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

FIG. 8 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

Brief Description of the QueryBook Platform Drawings (FIG. 9–FIG. 30)

FIG. 9 is a diagram of the queryBook system architecture (Prime Directive as Domain 0).

FIG. 10 is a diagram of the fact Unit — atom, envelope, and content-derived identity.

FIG. 11 is a diagram of the fact-certification authority hierarchy.

FIG. 12 is a diagram of the uFCS substrate — three-store architecture.

FIG. 13 is a diagram of the authored tour and exhibition sequencing.

FIG. 14 is a diagram of the fQL — joint structure and trust execution.

FIG. 15 is a diagram of the verifying machine-generated text against certified records.

FIG. 16 is a diagram of the federation — cross-boundary re-validation.

FIG. 17 is a diagram of the fact-level erasure by external-key destruction.

FIG. 18 is a diagram of the universal security architecture.

FIG. 19 is a diagram of the universal honeypot saturation and data firehose.

FIG. 20 is a diagram of the deployment topologies over an invariant common layer.

FIG. 21 is a diagram of the adaptive response engine — directive fixed before generation.

FIG. 22 is a diagram of the user-owned agent system with governance and kill switch.

FIG. 23 is a diagram of the deterministic multi-language regeneration.

FIG. 24 is a diagram of the semantic compression, expansion, colorization and upscaling.

FIG. 25 is a diagram of the queryBook-enhanced TCP/IP hybrid protocol packet.

FIG. 26 is a diagram of the neuromorphic cognitive layer — attractor convergence.

FIG. 27 is a diagram of the fact Unit — confidence and evidence model.

FIG. 28 is a diagram of the uFCS packet structure — serialized Fact Unit record.

FIG. 29 is a diagram of the uFCS packet framing — content-addressed block storage.

FIG. 30 is a diagram of the fQL — query anatomy and worked example.

Reference Numerals

No.	Element
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100	QueryBook LEL system
110	Sub-Language Priming Layer (SLPL)
120	Language Expression Layer (LEL)
130	Transition Gate
140	Phase Two semantic-communicative learning
150	Multilingual Delta Acquisition module
160	Speech Output Pipeline
165	natural speech output
170	co-grounding interface
180	Federated Query Language (FQL)
190	Universal Federated Connectivity System (UFCS)
195	federated data sources
200	SLPL
210	acoustic phonetic processing
211	signal-level feature embedding
220	phonics / phonemic awareness
221	phoneme-boundary segmenter
230	communication-science foundations
231	communicative-salience weight
240	prosody & rhythm processing
241	prosodic contour vector
250	information-theoretic grounding
251	information weighting
260	neurolinguistic chunking
261	chunk segments
270	licensed multimedia input
300	developmental learning sequence
310	Phase One (structural-acoustic)
311	sound discrimination
312	syllabic segmentation
313	word-as-acoustic-object
314	semantics suppressed
320	Transition Gate (one-way)
321	phonemic-inventory completeness $\geq \tau_1$
322	lexical stability $\geq \tau_2$
323	co-occurrence density $\geq \tau_3$
324	pattern saturation $< \epsilon$
330	Phase Two (semantic-communicative)
331	word-to-referent grounding
332	communicative discovery
333	emergent grammar / syntax
400	Delta Acquisition Model
410	reused L1 foundation
420	L2 delta detector
430	targeted training of novel elements
440	multilingual semantic alignment
450	acquisition acceleration flywheel
451	negative-transfer detection & retraining
500	Speech Output Pipeline
510	lexical selection
520	grapheme-to-phoneme conversion
530	coarticulation modeling
540	prosody generation
550	register & affect calibration
560	self-monitoring feedback loop
570	neural vocoder rendering

No.	Element
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950	syllable count + primary-stress derivation
960	explicit null result (not fabricated)
970	phrase-level single-call transcription
971	displayed transcription consistent with synthesized audio
980	self-contained application bundle
981	bundled open-source speech engine
982	runtime-support libraries
983	bundled phoneme data
984	engine-selection ladder (per OS / language)
985	host OS built-in voice service
986	culture-matched OS voice (when installed)
987	honest refusal with remedy (no fabricated audio)
990	deterministic translator (through English)
991	bundled open-source bilingual dictionary
992	coverage accounting + unknown-word passthrough (flagged)
993	deterministic language detector
995	permanent quality-control battery
996	IPA-produced check
997	syllable-sanity check (± 1)
998	valid-waveform synthesis check
999	per-language QC report + overall pass score
1000	LIL voiceprint & identity subsystem
1001	consent gate (precondition)
1002	audio/video capture + VAD/segmentation
1003	acoustic feature extraction
1004	speaker embedding (voiceprint vector)
1005	enrollment — canonical voiceprint
1006	match / score (cosine · PLDA)
1007	QueryBook trust gate (VERIFIED/CONTRADICTED/UNKNOWN)
1008	Fact-Unit identity graph
1009	LEL cross-script name/entity resolution
1010	provenance-tracked link Fact Unit
1020	voice authentication request
1021	capture phrase + embed
1022	anti-spoofing (liveness/replay/synthetic)
1023	deterministic risk-scoring policy
1024	authentication decision gate
1025	second-factor challenge
1026	auth-event Fact Unit (provenance)
1100	UIAS integrity audit service
1101	monitored system workflow (unmodified)
1102	input-boundary position
1103	processing-layer position
1104	storage-layer position
1105	output-boundary / handoff position
1106	Connector + Ingestion agents (seal at observation)
1107	seal = Fact Unit (hash · id-hash · independent timestamp · source · agent)
1108	Similarity Detection Agent (semantic signatures)
1109	Fact-Unit audit ledger (UFCS, append-only, immutable)
1110	contemporaneous cross-party anomaly flag
1111	Verification Agent (continuous re-verification)
1112	content-not-stored / one-way identity (fact-level erasure)
1120	UIAS signed Proof Package generation
1121	accusation / inquiry
1122	Proof Package Agent
1123	sealed Fact Units + verified seals

No.	Element	No.	Element
580	natural speech waveform	1124	cross-submission similarity log entry
600	co-grounded integration	1125	chronological ledger extract (RPH)
610	natural-language input	1126	Arbitration Agent (factual determination; Prime-Directive gate)
620	LEL analysis	1127	assemble + sign under UIAS independent identity
630	FQL operations	1128	signed Proof Package
640	UFCS execution	1129	factual-only determination (legal reserved to courts/regulators)
650	federated sources	1200	PIL paralinguistic & pragmatic interpretation
660	results	1201	ingested voice (consent-gated)
670	render to NL / speech in input language	1202	transcription (words)
680	language-agnostic resolution	1203	Paralinguistic Analyzer (measured)
700	Language Lab prototype embodiment	1204	tone/loudness/pacing/emphasis + labeled arousal estimate
710	corpus source	1205	Pragmatic Framer (sourced)
720	Phase-One structural agent	1206	context · perspective · ergonomics · culture · social mores
730	structural analyzer	1207	Interpretation Frame (labeled estimates + sourced frames)
731	grapheme inventory	1208	Fact Units on ingested content
732	rule-seeded G2P	1209	covenant: estimates not asserted; frames sourced; consent
733	syllabifier	1210	PIL human query interpretation
734	lexical-stability metric	1211	human query (typed/spoken)
735	co-occurrence density metric	1212	Interpretation Frame (para + pragmatic)
736	prosody proxy	1213	intent disambiguation (context/perspective)
740	Fact Unit store	1214	ambiguity gate
750	Transition-Gate metrics	1215	ask one clarifying question / stop
751	one-way latch	1216	register & ergonomic mode selector
760	phase agents on shared manager	1217	culture/mores/ergonomics packs
761	Phase-1 agent (implemented)	1218	retrieve-gate-compose (verified Fact Units; cite or refuse)
762	Phase-2 agent — deterministic dictionary + store grounding (implemented)	1219	answer in selected register/mode (facts unchanged)
763	Phase-3 agent (declines — spec-only)	1300	Lecture Query ingestion pipeline
764	Phase-4 agent (declines — spec-only)	1301	live or recorded audio/video program
800	deterministic Phase-Two semantic grounding	1302	capture + consent gate
810	Phase-One vocabulary (learned words)	1303	ASR transcription + diarization (untrusted input)
820	deterministic grounder (per word)	1304	video frames → faces (LIL)
830	bundled public-domain dictionary source	1305	PIL analysis over time (tone/pacing/arousal + frame)
831	dictionary meaning Fact Unit (means, trust 0.9)	1306	LEL segment/structure/extract claims
840	self-grounding store-lookup source	1307	deterministic summary (TOC/key points/claims)
841	store-link Fact Unit (grounded_by_fact)	1308	ingest → Lecture Record; claims = attributed utterance facts
850	Fact Unit store (provenance, domain language)	1309	forever-link: speaker/venue/event/org/date (immutable, trust-gated)
860	excluded large-language-model meaning source	1310	forever-linked Lecture Record (queryable)
870	optional future dictionary-gated suggestor	1311	Lecture Record (content-addressed)
900	advanced multilingual pronunciation subsystem	1312	speaker entity (LIL voiceprint/face)
910	pronunciation request (word/phrase, language)	1313	venue entity
920	pronunciation preference-ladder selector	1314	event entity
930	bundled open-source phonemizer	1315	organization + date
931	phonemizer IPA transcription	1316	transcript + diarization
932	phonemizer call cache	1317	PIL track (tone/pacing over time)
933	bounded call log / timeout guard	1318	attributed claims → corroborated/contradicted/open
940	rule-seeded G2P tables (per language)	1319	FQL / Q&A over the record
941	approximate rule-seeded IPA (offline)	1320	answer: cited, attributed, refused where unknown

Drawings

Formal patent drawings — black line art, forming **one QueryBook figure set**: FIG. 1–FIG. 8 and FIG. 31–FIG. 32 depict QueryBook’s integral developmental language and speech subsystem (the Language Expression Layer), FIG. 9–FIG. 30 depict the QueryBook platform in which that subsystem is co-grounded (see FIG. 6), FIG. 33–FIG. 34 depict the Language & Identity Linking (LIL) voiceprint and multimodal identity subsystem, FIG. 35–FIG. 36 depict the Universal Integrity Audit Service (UIAS) that QueryBook delivers on these primitives, FIG. 37–FIG. 38 depict the Paralinguistic & Pragmatic Interpretation Layer (PIL), and FIG. 39–FIG. 40 depict the Lecture Query interaction mode. Each figure is on its own sheet and every element is identified by a reference numeral used consistently in the Brief Description of the Drawings, the Reference Numerals key, and the Detailed Description. Figures are draft; counsel/draftsperson to finalize to USPTO drawing standards (37 CFR 1.84) before filing.

FIG. 1

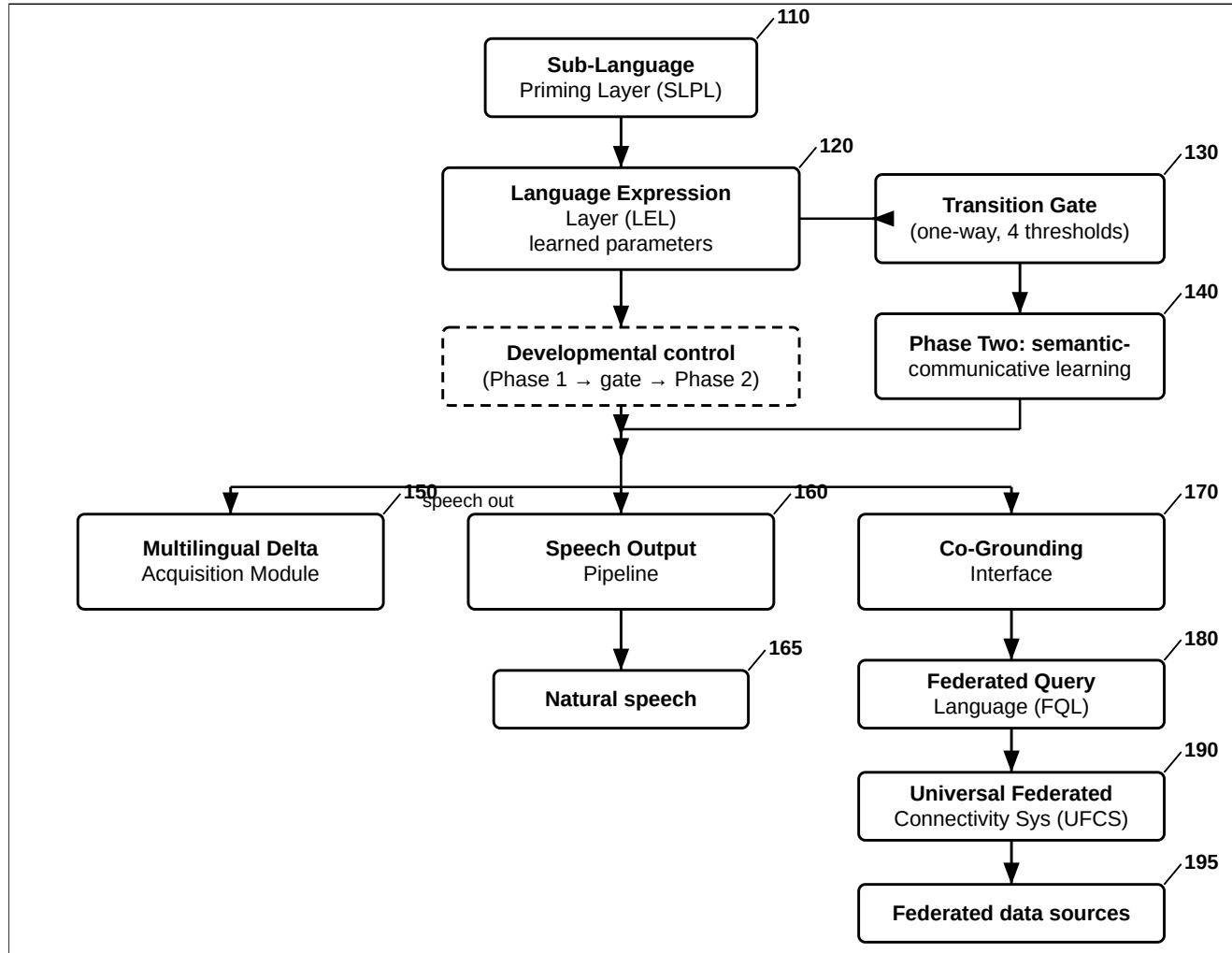


FIG. 1 — QueryBook LEL — Overall System Architecture

FIG. 2

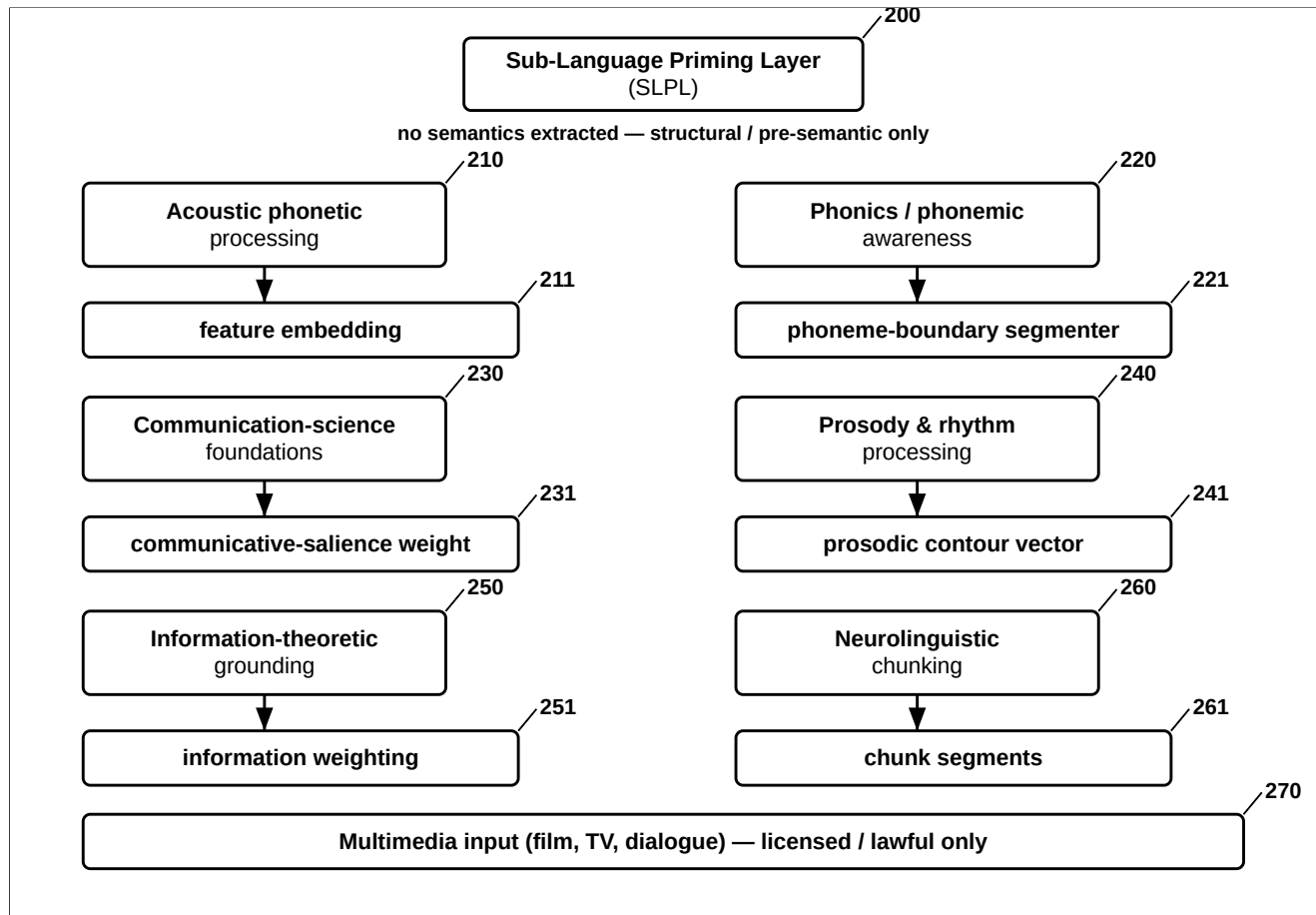


FIG. 2 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 3

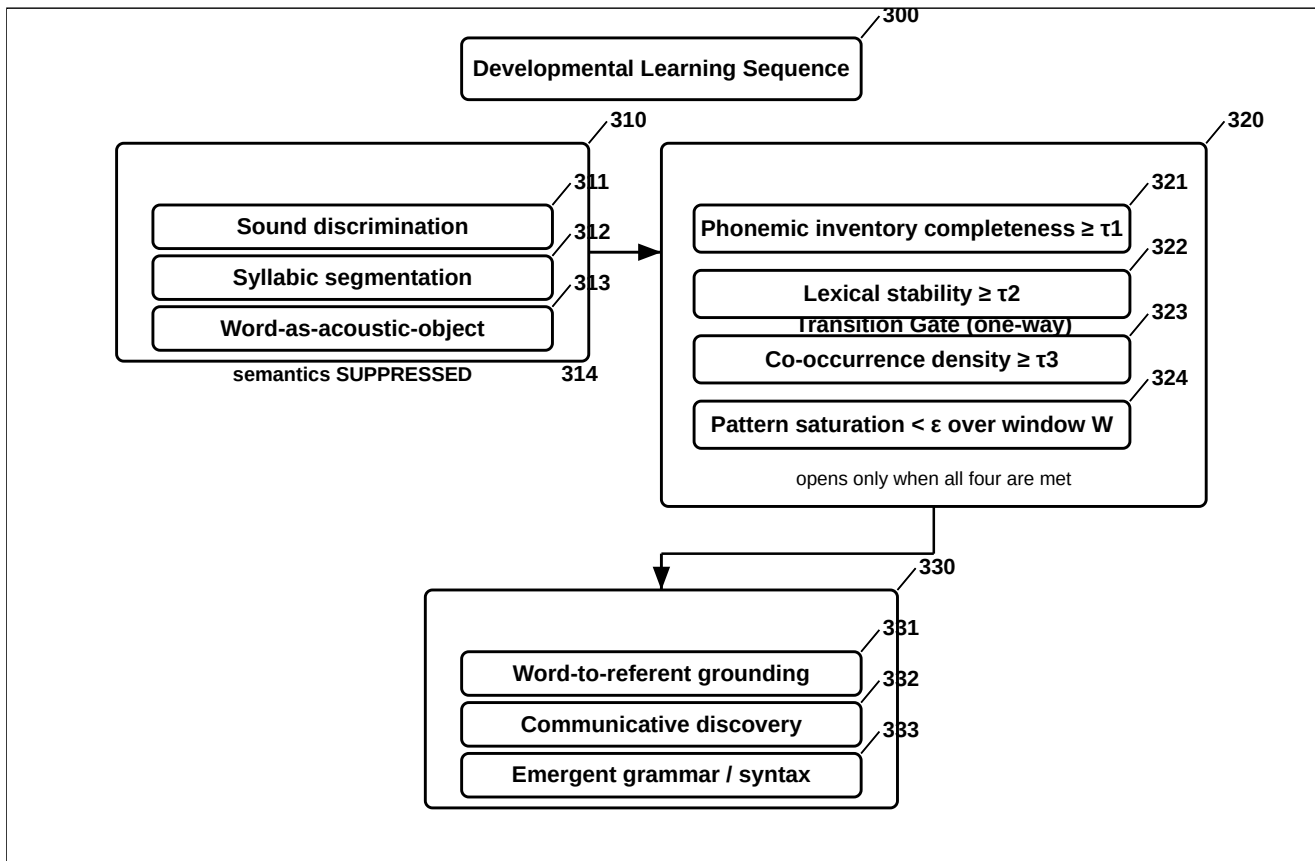


FIG. 3 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 4

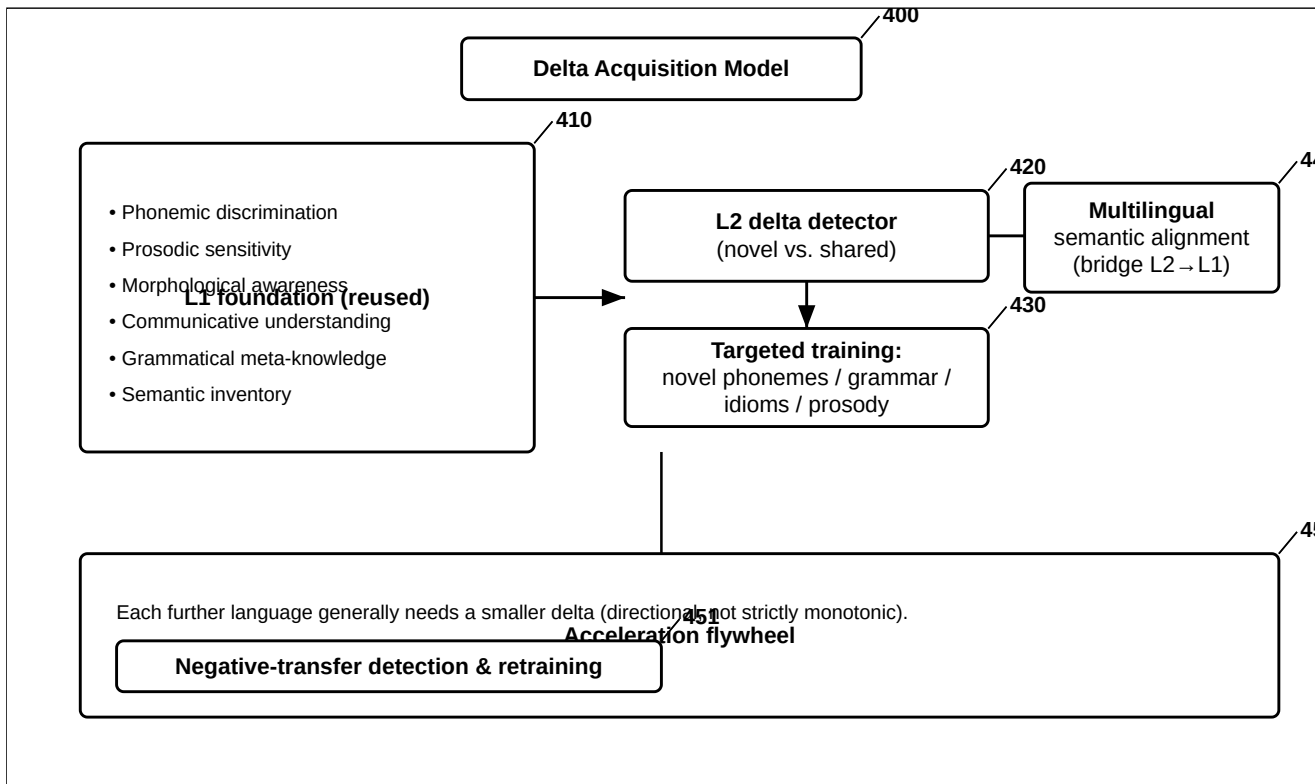


FIG. 4 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 5

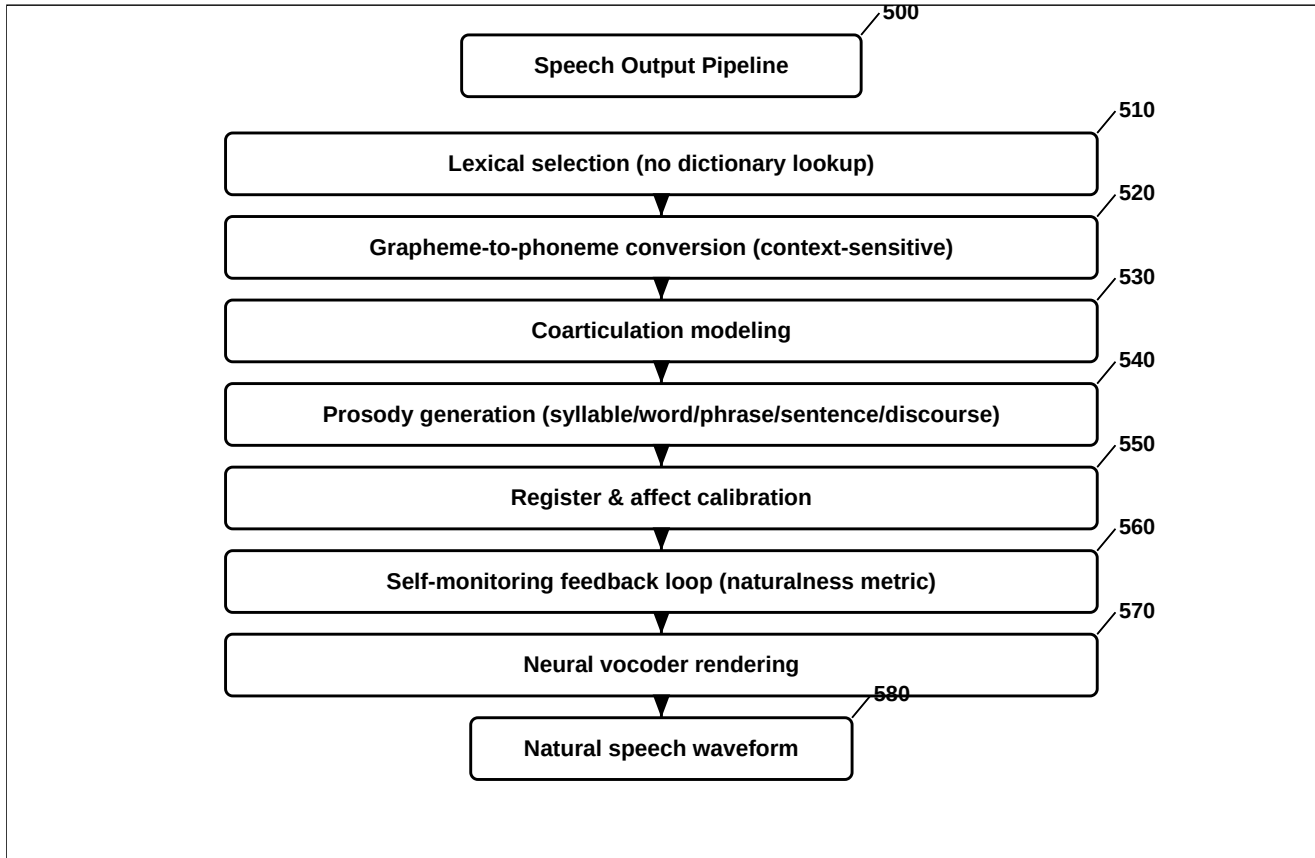


FIG. 5 — Speech Output Pipeline (Seven Learned Stages)

FIG. 6

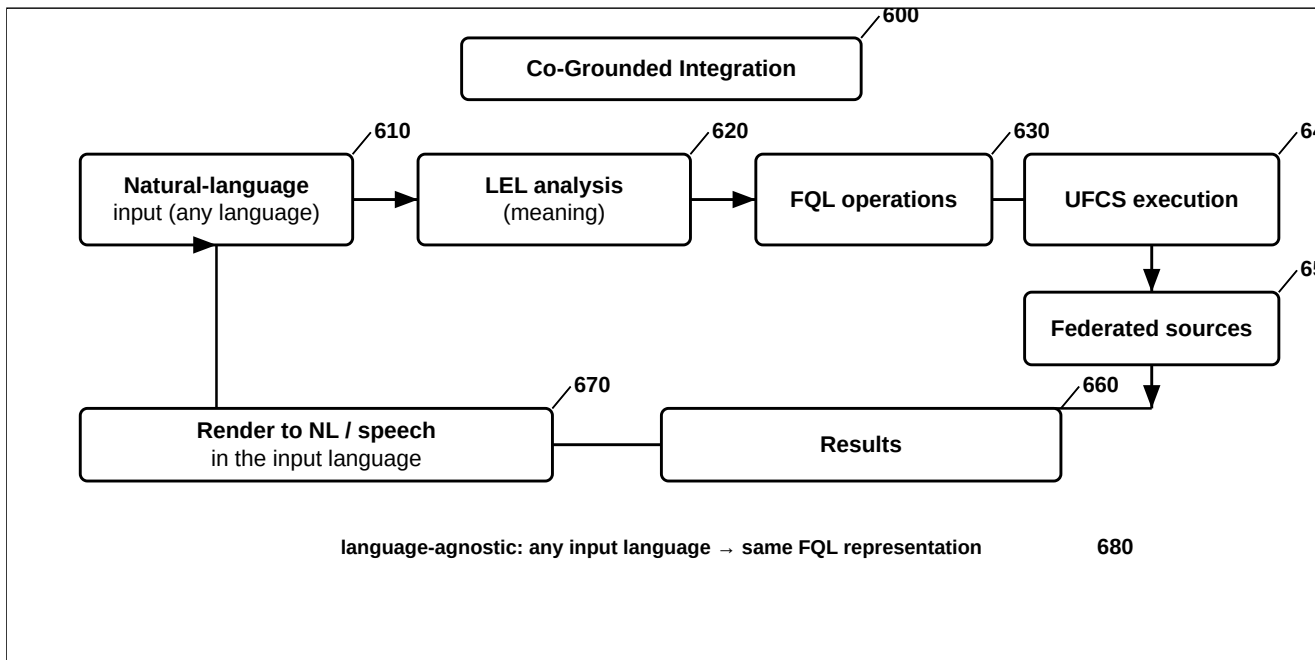


FIG. 6 — Co-Grounded Integration with FQL and UFCS

FIG. 7

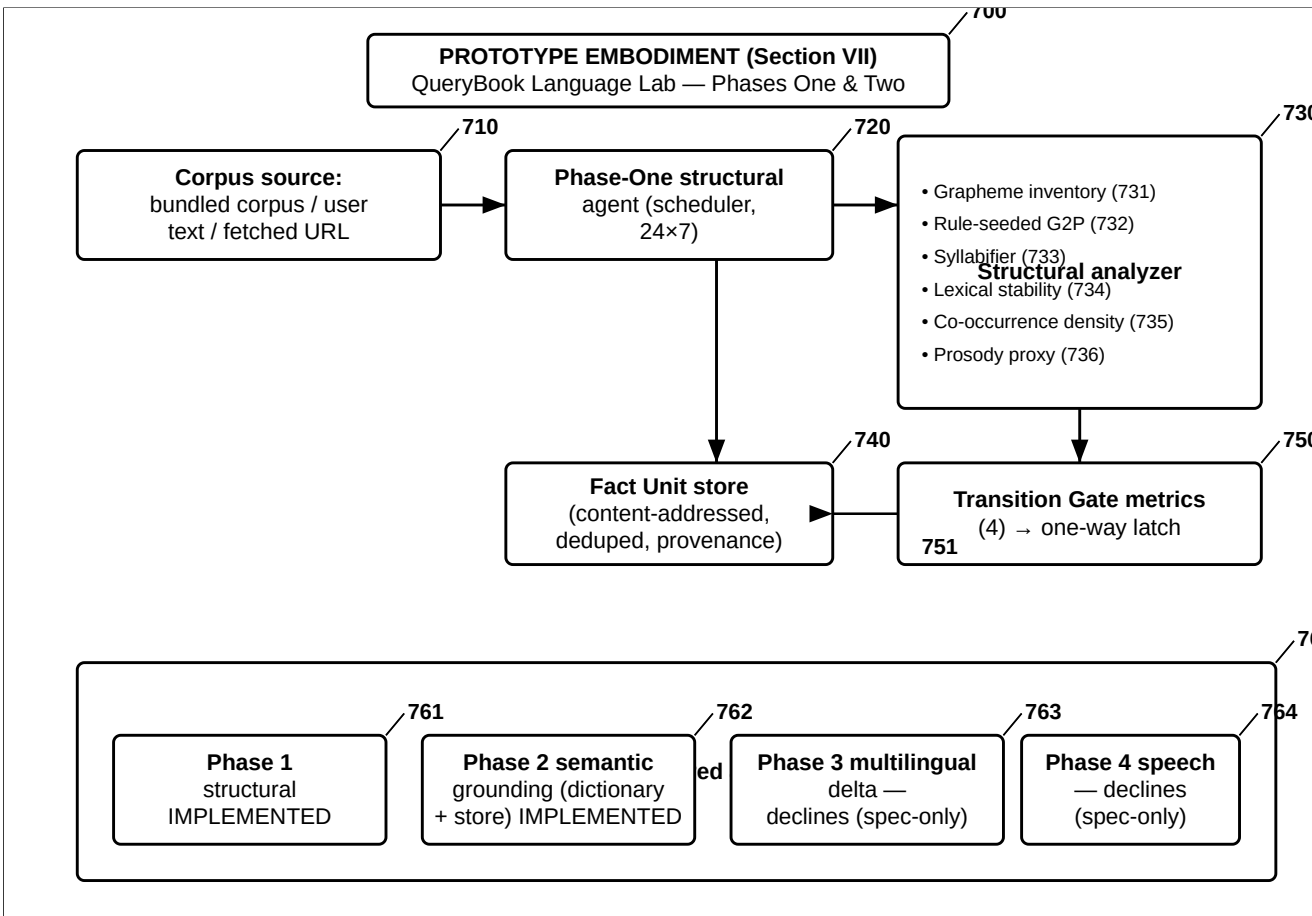


FIG. 7 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 8

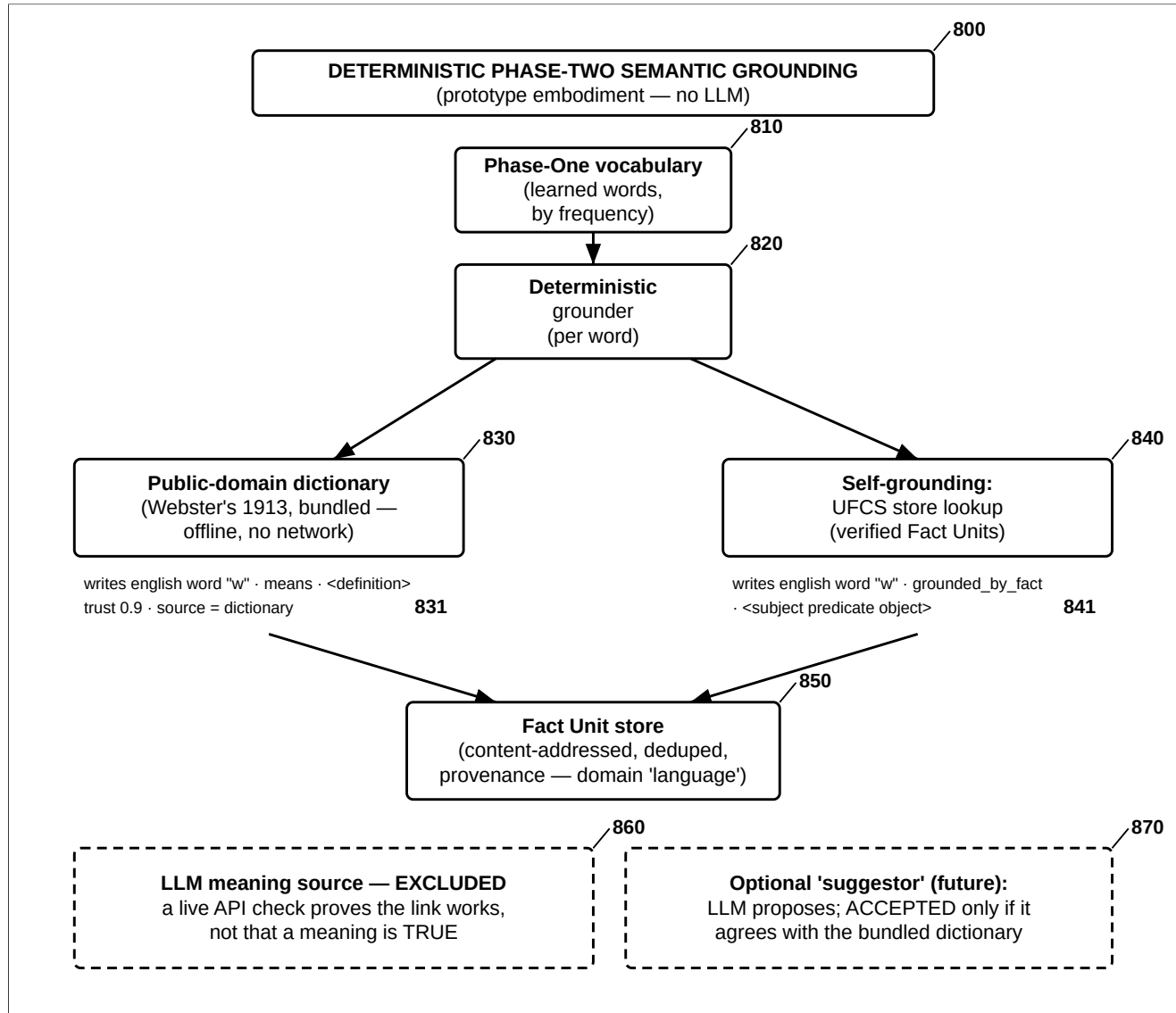


FIG. 8 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

QueryBook Platform Drawings (FIG. 9–FIG. 30) — the system in which the language subsystem is integral

FIG. 9 through FIG. 30 depict the QueryBook deterministic, provenance-tracked federated knowledge and query platform of which the developmental language and speech subsystem (the Language Expression Layer, FIG. 1–FIG. 8 and FIG. 31–FIG. 32) is an integral, co-grounded part (see FIG. 6). Together with the language-subsystem figures these form **one QueryBook figure set**. Draft — the language-subsystem figures use a 100–999 numeral series and the platform figures a 100–2200 series; counsel/draftsperson to consolidate the two into a single QueryBook numeral series and finalize to 37 CFR 1.84 before filing.

FIG. 9

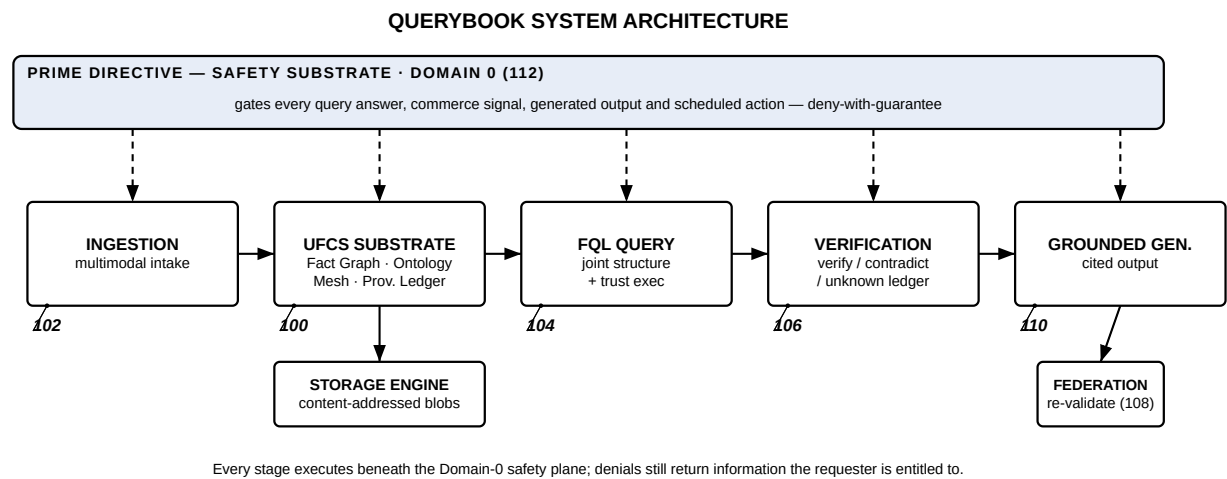


FIG. 9

FIG. 9 — QueryBook system architecture (Prime Directive as Domain 0)

FIG. 10

FACT UNIT — ATOM, ENVELOPE, AND CONTENT-DERIVED IDENTITY

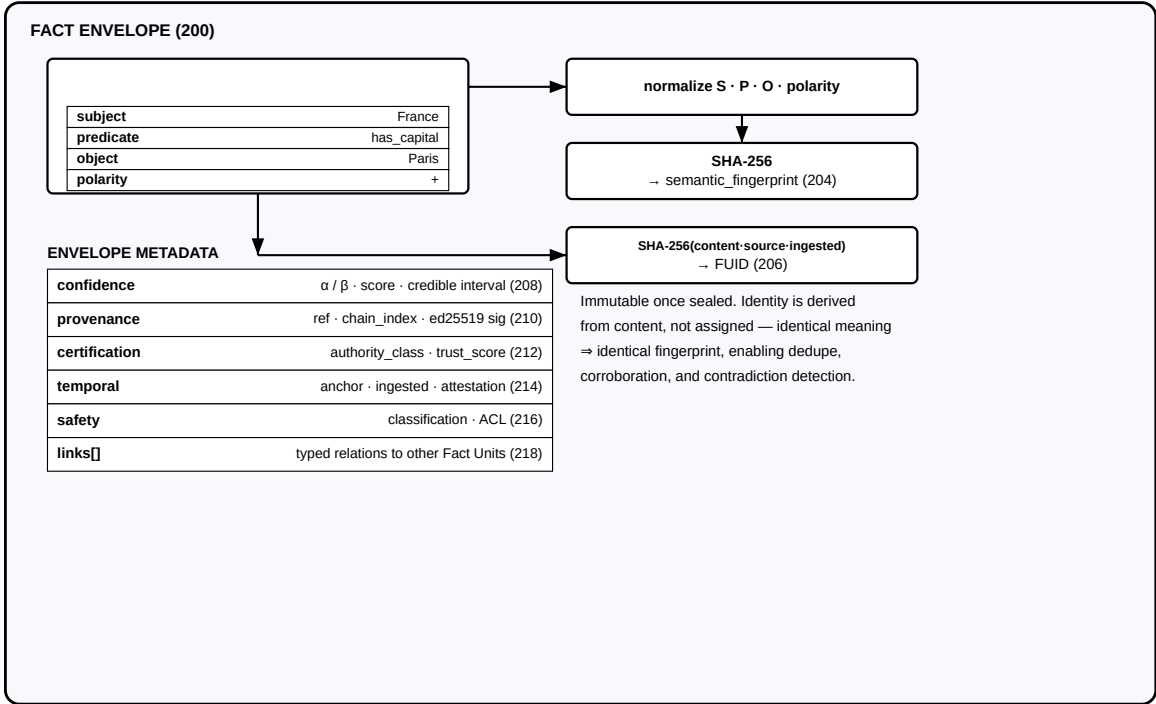


FIG. 10

FIG. 10 — Fact Unit — atom, envelope, and content-derived identity

FIG. 11

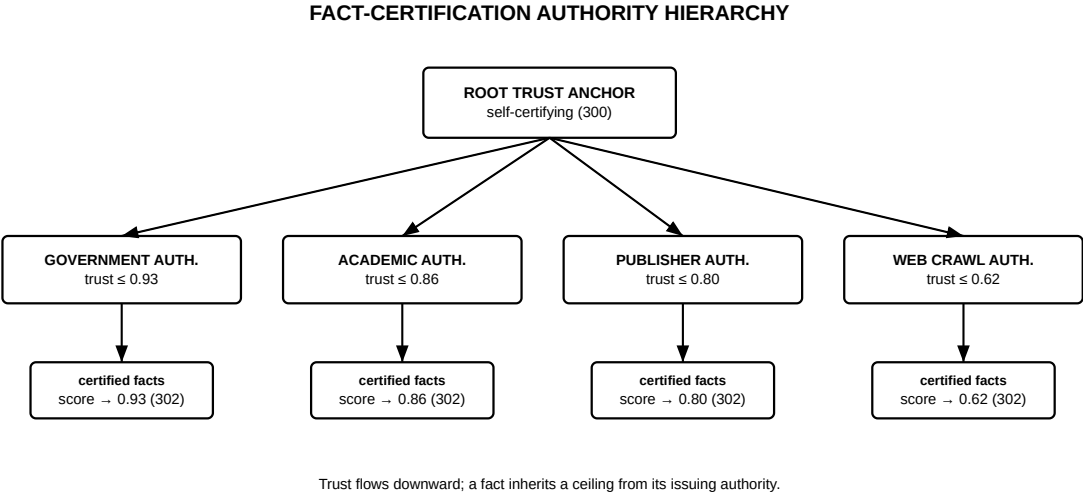


FIG. 11

FIG. 11 — Fact-certification authority hierarchy

FIG. 12

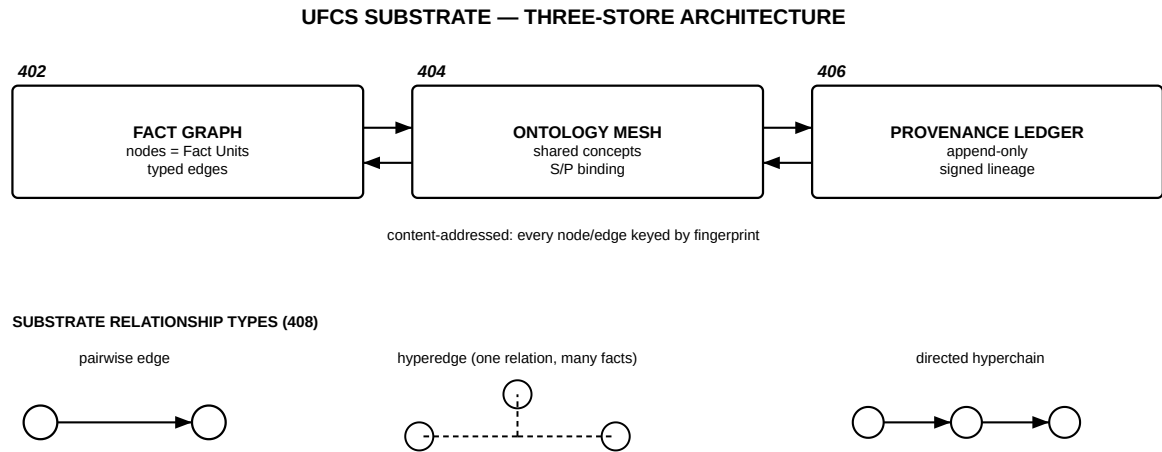
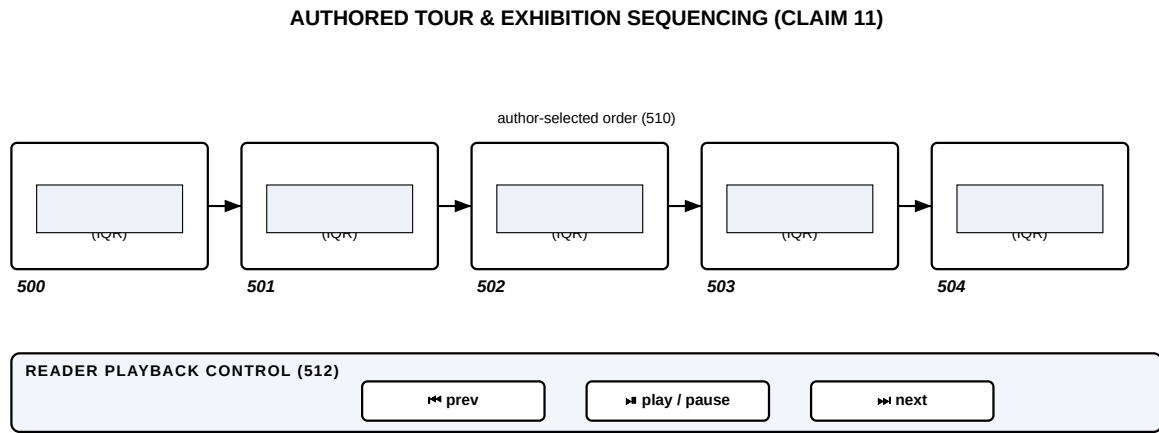


FIG. 12

FIG. 12 — UFCS substrate — three-store architecture

FIG. 13



Author fixes the sequence; the reader controls pace and direction along the ordered waypoints.

FIG. 13

FIG. 13 — Authored tour and exhibition sequencing

FIG. 14

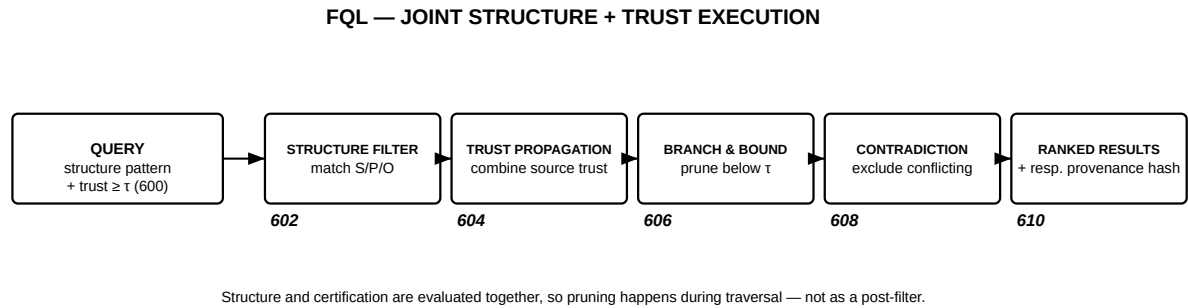


FIG. 14

FIG. 14 — FQL — joint structure and trust execution

FIG. 15

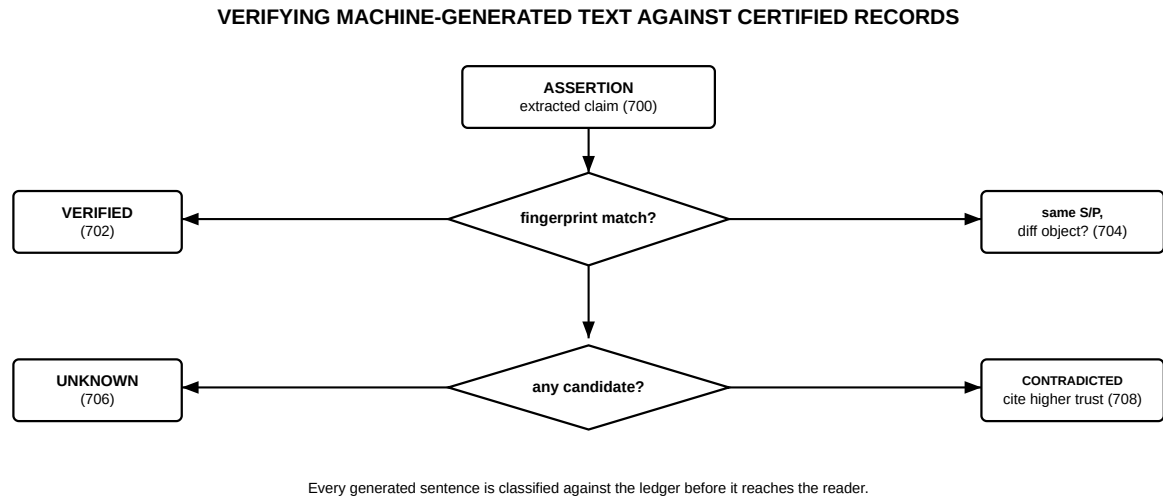


FIG. 15

FIG. 15 — Verifying machine-generated text against certified records

FIG. 16

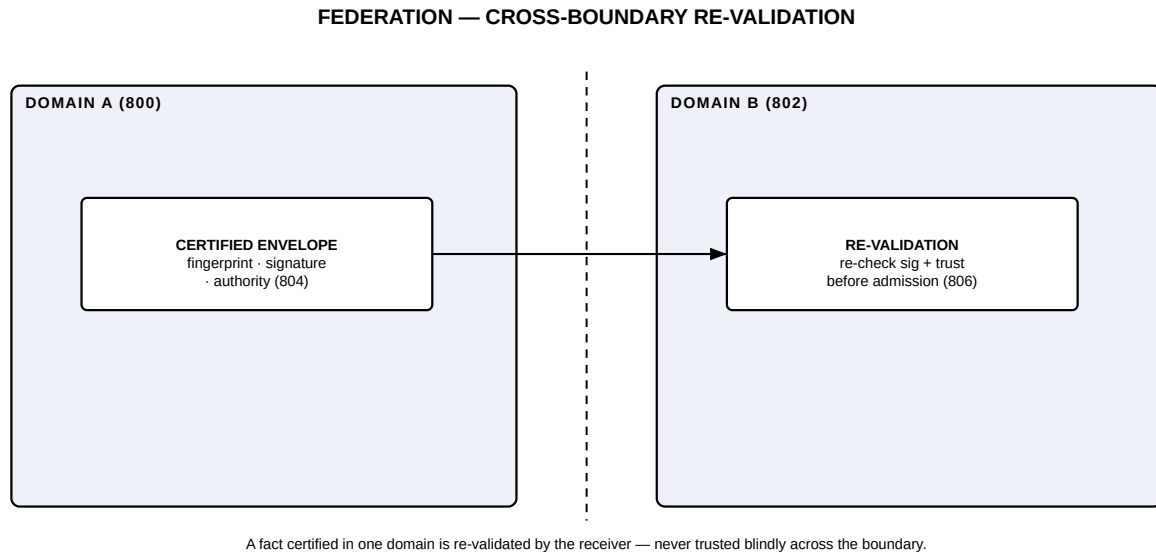
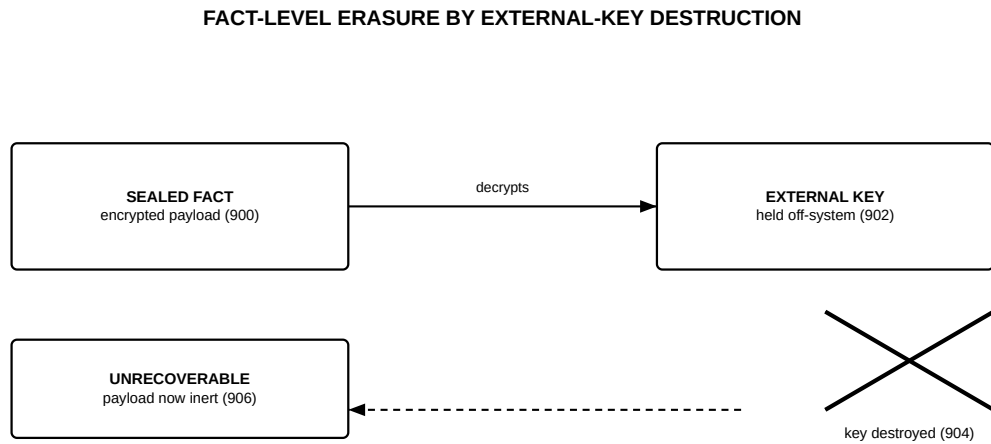


FIG. 16

FIG. 16 — Federation — cross-boundary re-validation

FIG. 17



Erase one fact by destroying its key; the content-addressed store is otherwise append-only and immutable.

FIG. 17

FIG. 17 — Fact-level erasure by external-key destruction

FIG. 18

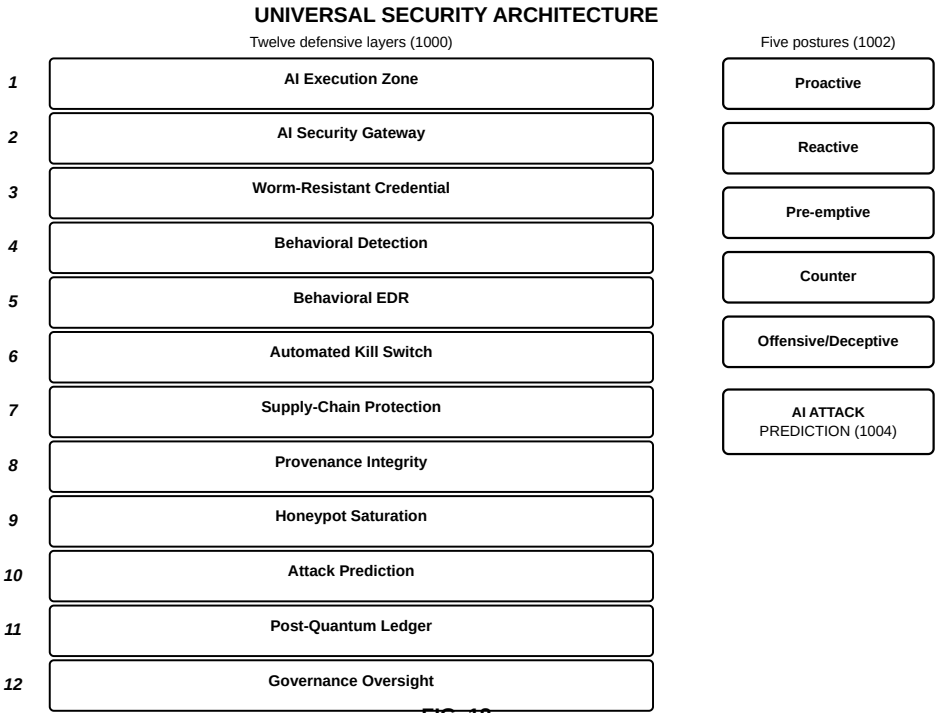


FIG. 18 — Universal security architecture

FIG. 19

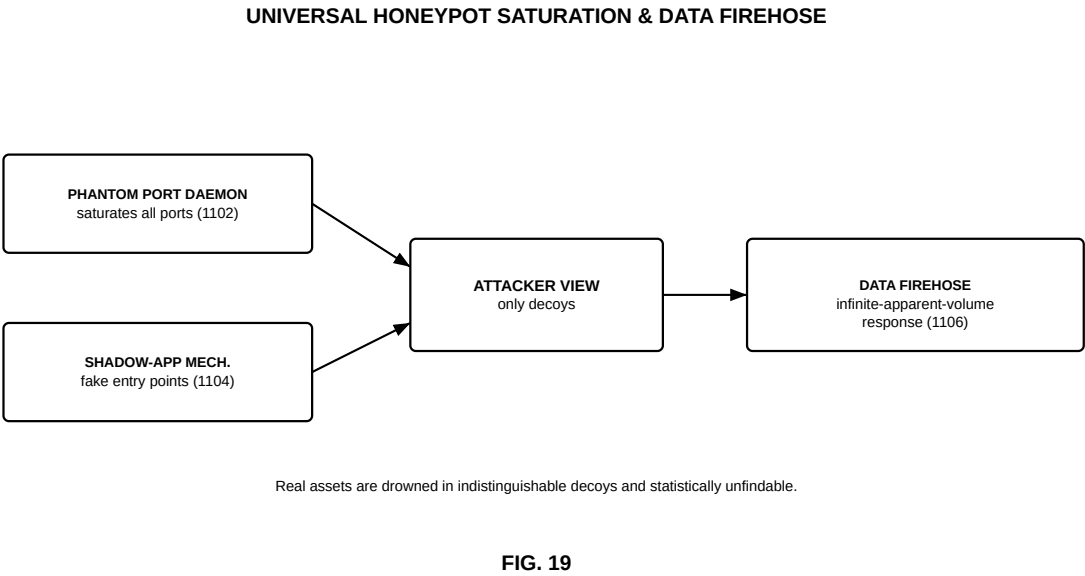


FIG. 19 — Universal honeypot saturation and data firehose

FIG. 20

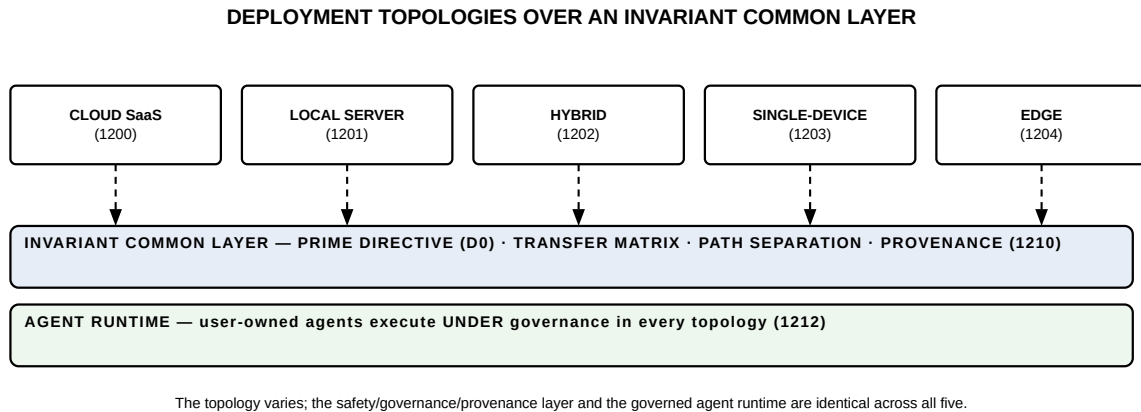


FIG. 20

FIG. 20 — Deployment topologies over an invariant common layer

FIG. 21

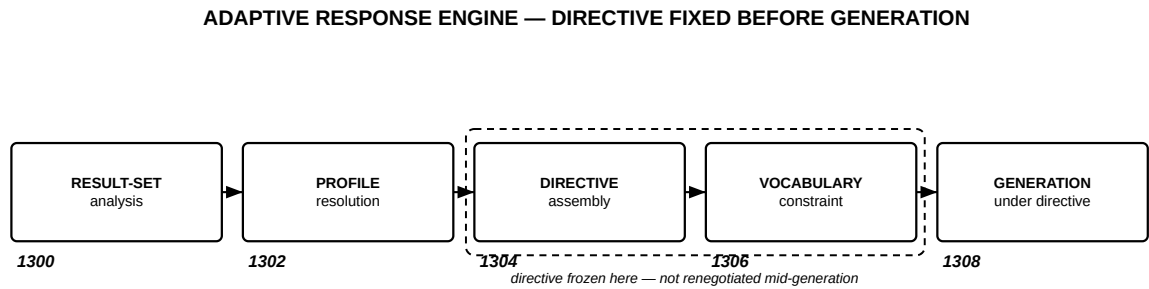
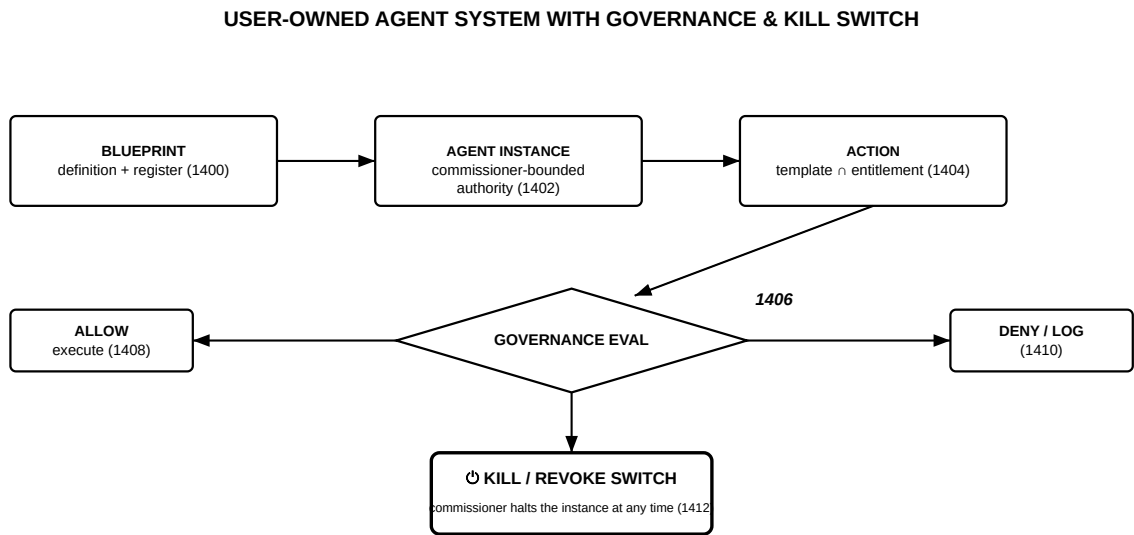


FIG. 21

FIG. 21 — Adaptive response engine — directive fixed before generation

FIG. 22

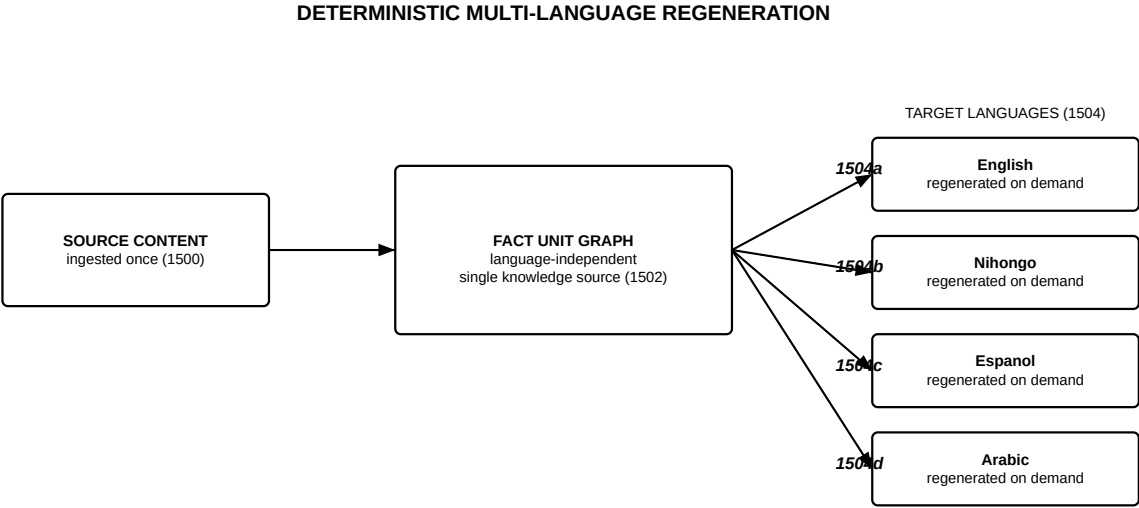


Governance is checked at every action; an agent can never exceed its commissioner, who holds an unconditional kill control.

FIG. 22

FIG. 22 — User-owned agent system with governance and kill switch

FIG. 23



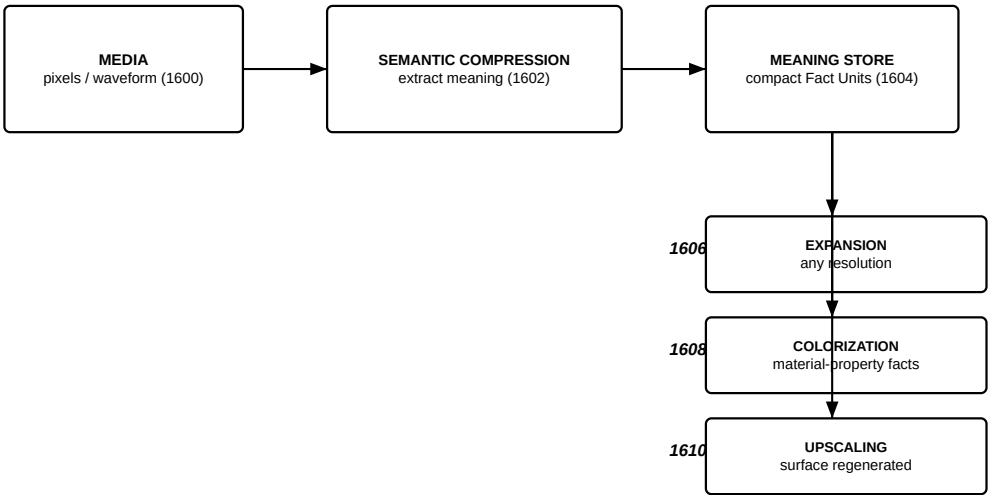
One graph, many surfaces: no per-language re-ingestion; each language is a projection of the same facts.

FIG. 23

FIG. 23 — Deterministic multi-language regeneration

FIG. 24

SEMANTIC COMPRESSION, EXPANSION, COLORIZATION & UPSCALING



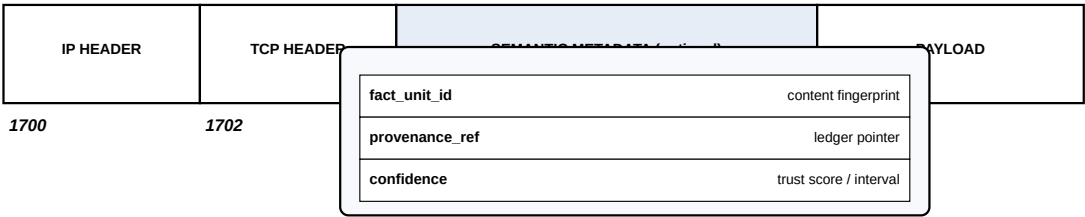
Store meaning, not surface; compression and regeneration are the same primitive — no generational quality loss.

FIG. 24

FIG. 24 — Semantic compression, expansion, colorization and upscaling

FIG. 25

QUERYBOOK-ENHANCED TCP/IP HYBRID PROTOCOL PACKET



A non-participating endpoint ignores the semantic header — provenance travels with the bits, backward-compatible.

FIG. 25

FIG. 25 — QueryBook-enhanced TCP/IP hybrid protocol packet

FIG. 26

NEUROMORPHIC COGNITIVE LAYER — ATTRACTOR CONVERGENCE

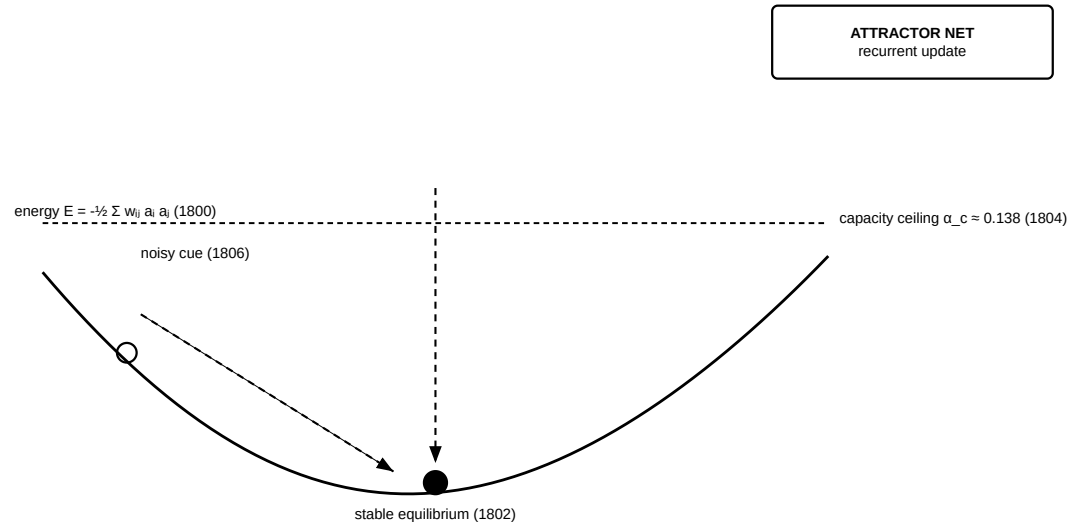
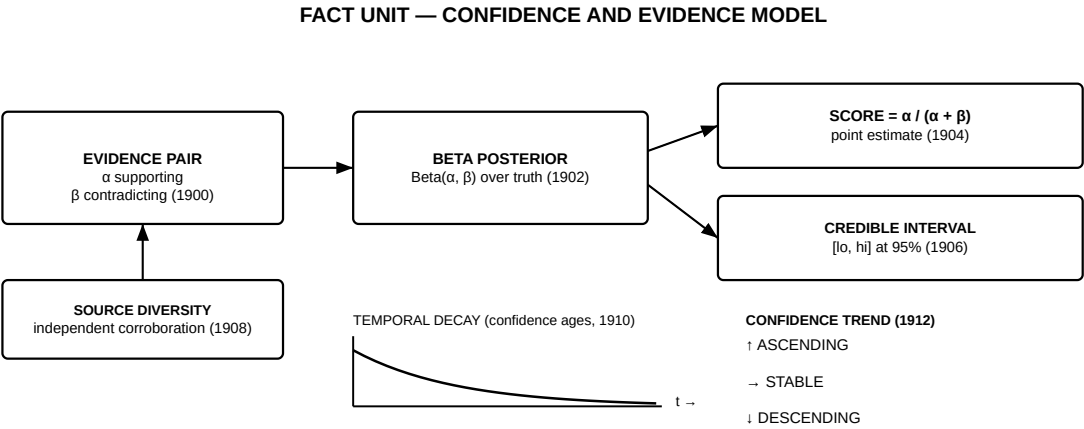


FIG. 26

FIG. 26 — Neuromorphic cognitive layer — attractor convergence

FIG. 27



Trust is a governed evidence estimate — not a single number: it carries an interval, ages over time, and reports its direction.

FIG. 27

FIG. 27 — Fact Unit — confidence and evidence model

FIG. 28

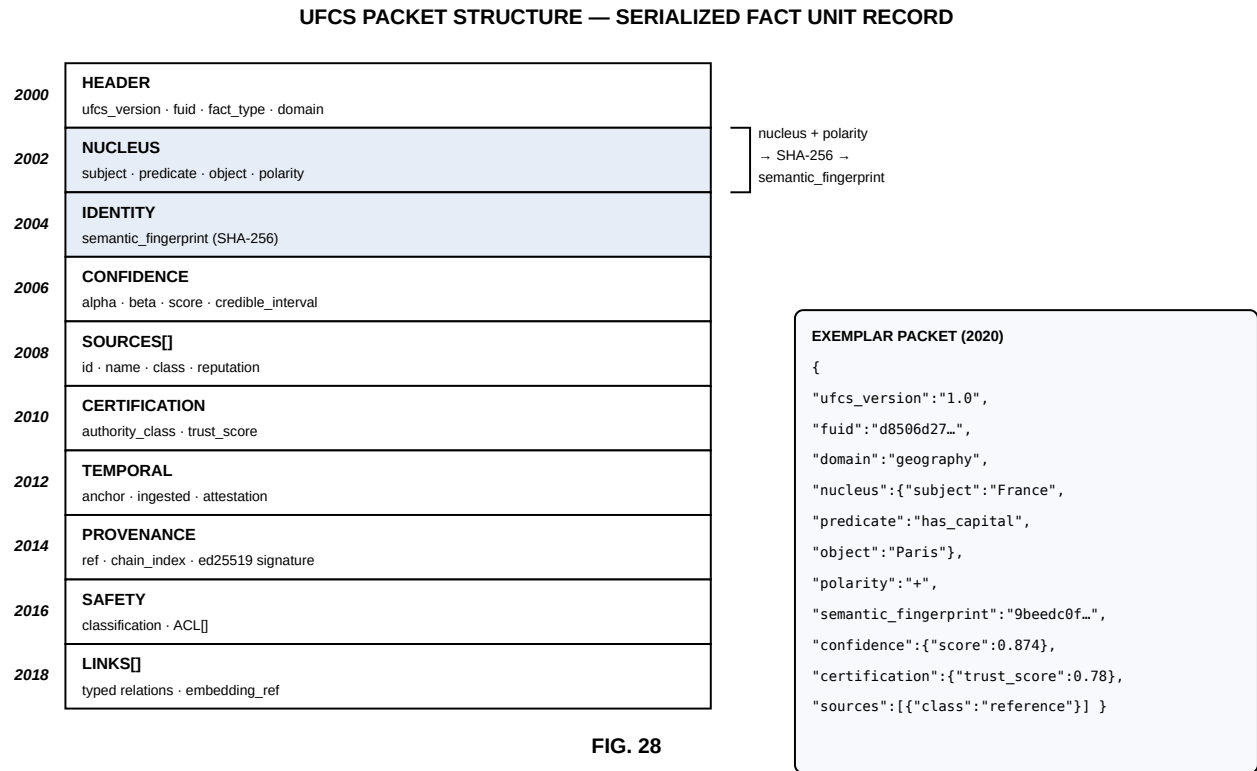


FIG. 28 — UFCS packet structure — serialized Fact Unit record

FIG. 29

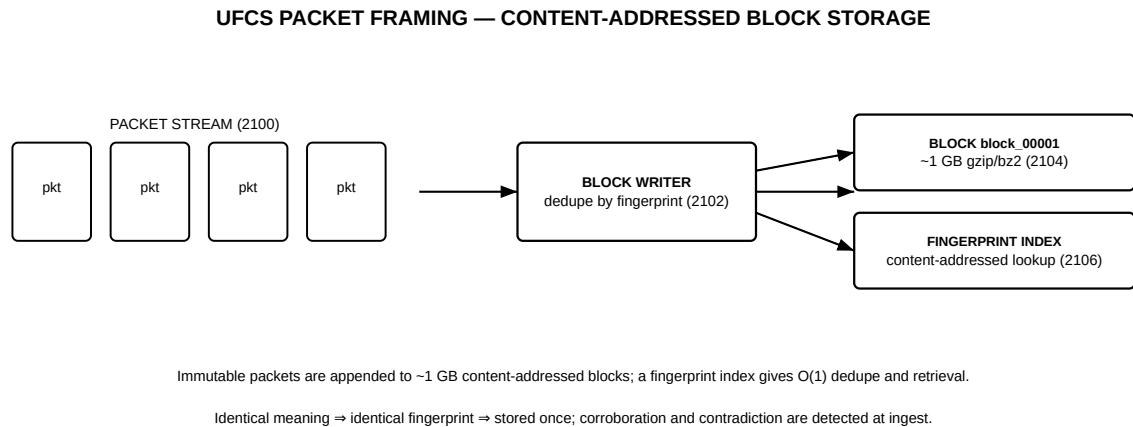


FIG. 29

FIG. 29 — UFCS packet framing — content-addressed block storage

FIG. 30

FQL — QUERY ANATOMY AND WORKED EXAMPLE

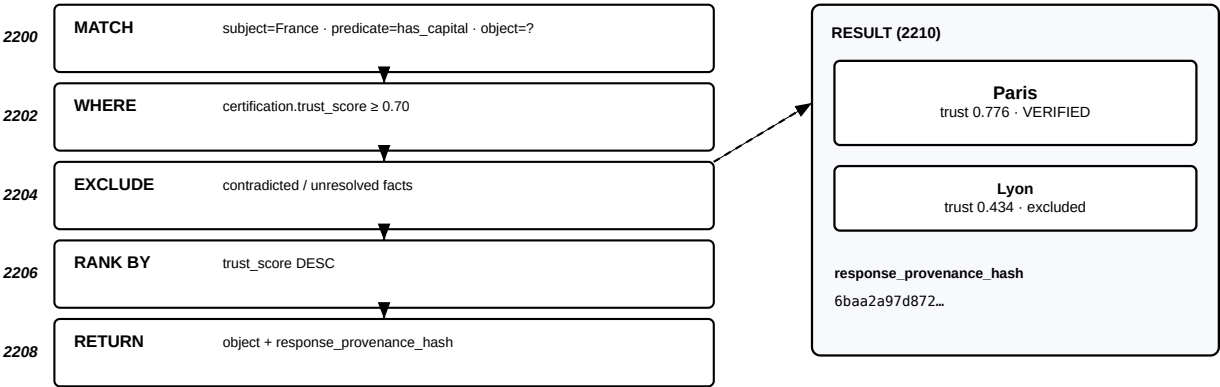


FIG. 30

FIG. 30 — FQL — query anatomy and worked example

FIG. 31

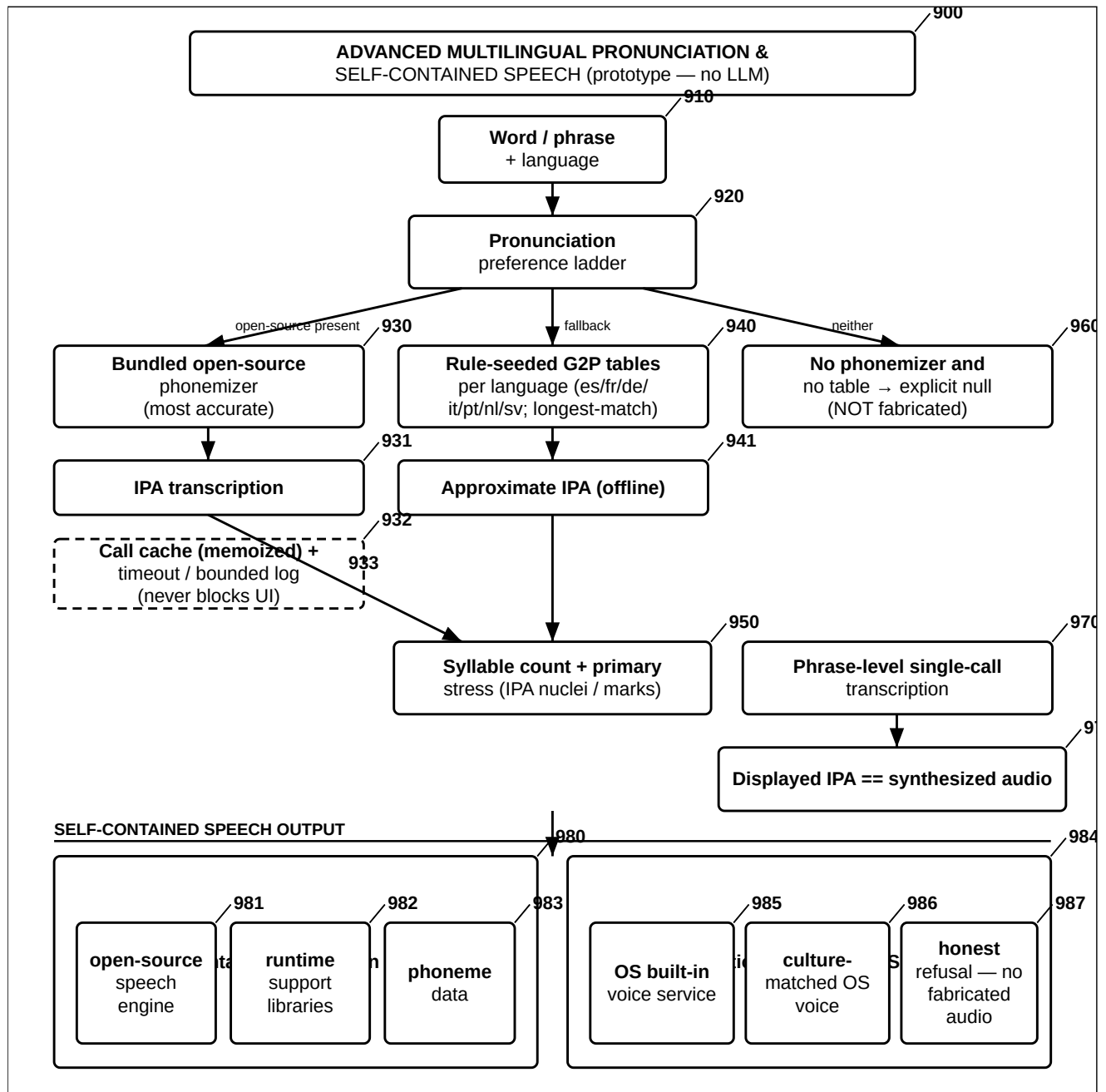


FIG. 31 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 32

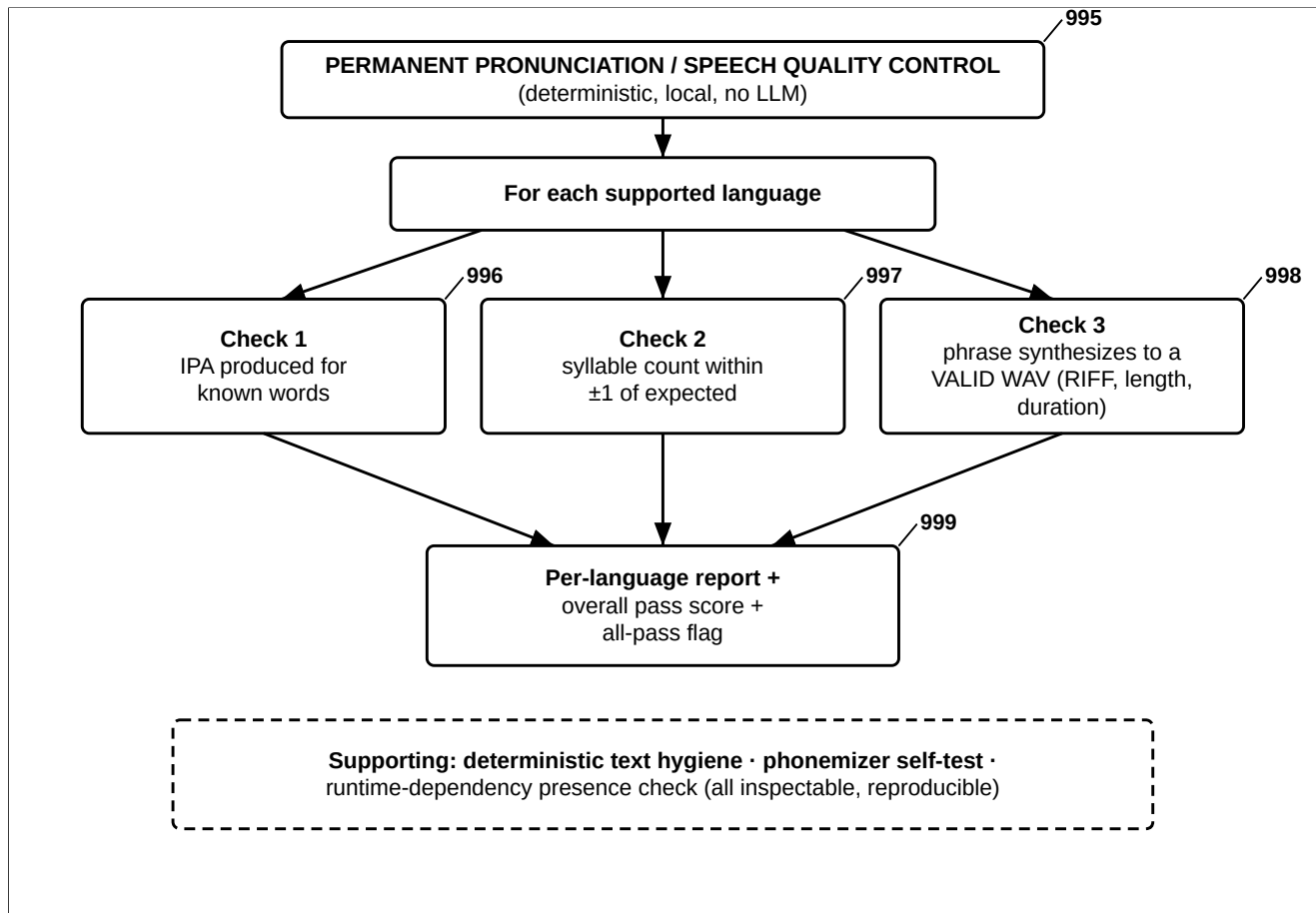


FIG. 32 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIG. 33

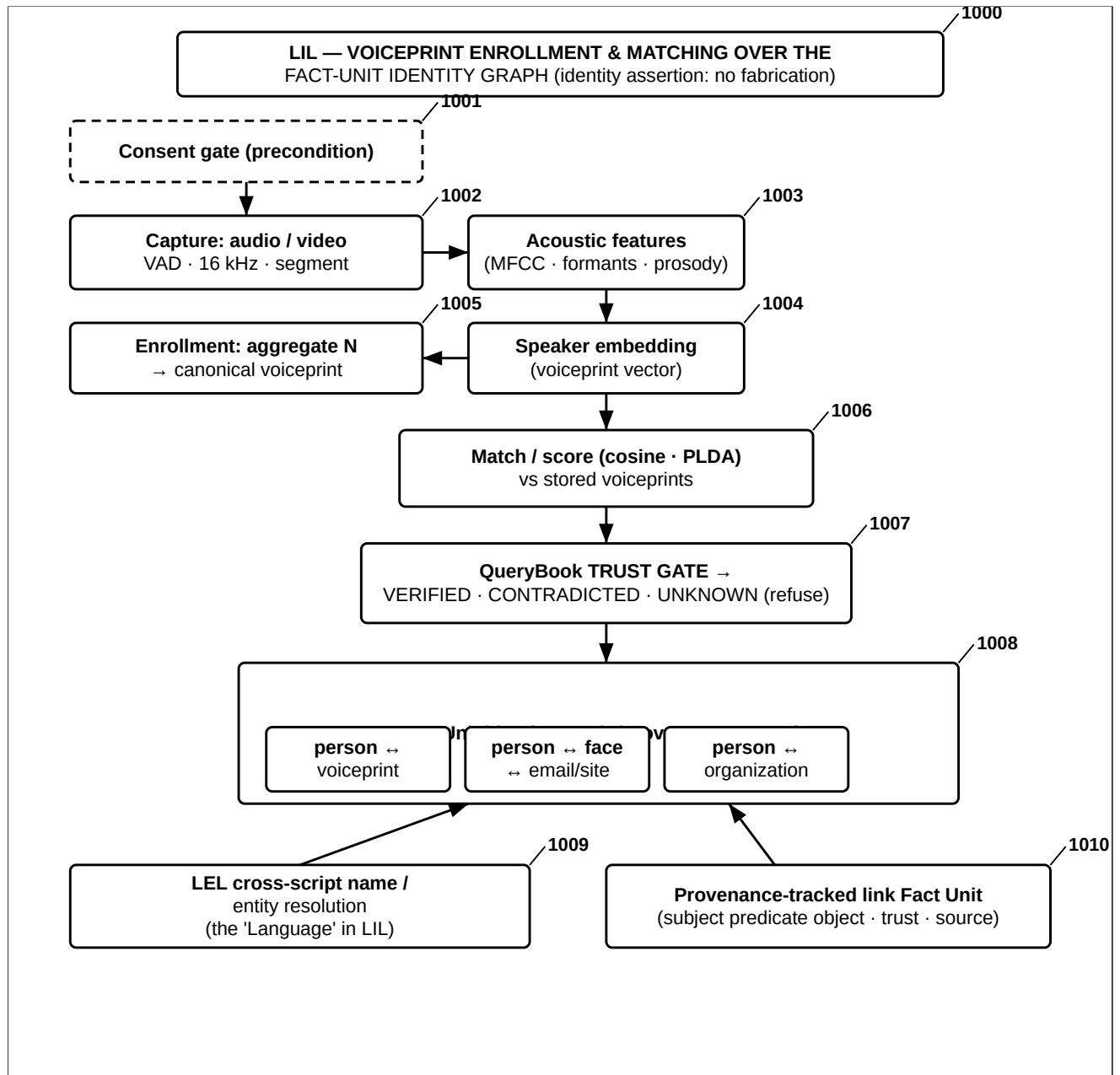


FIG. 33 — LIL — Voiceprint Enrollment and Matching over the Fact-Unit Identity Graph

FIG. 34

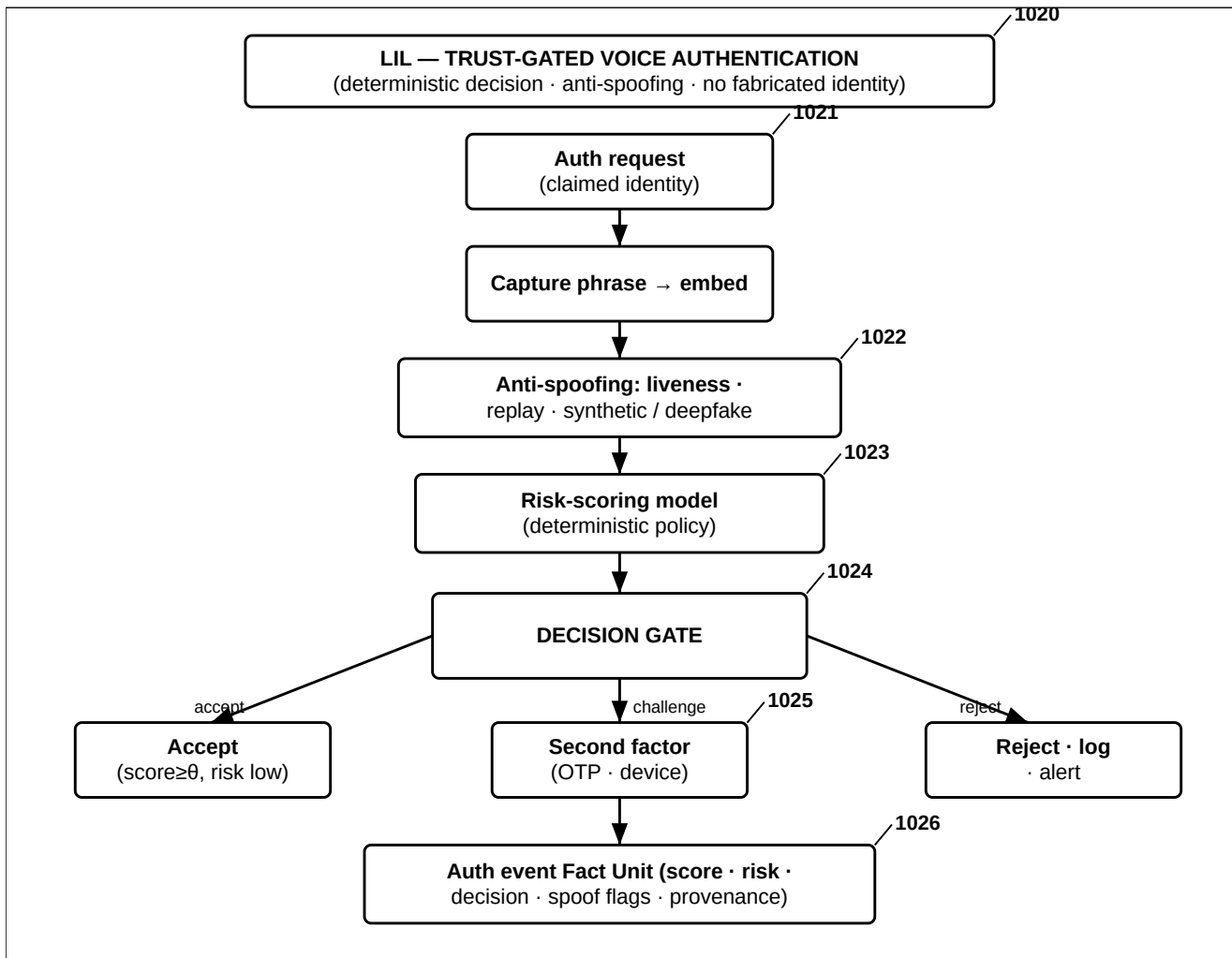


FIG. 34 — LIL — Trust-Gated Voice Authentication with Anti-Spoofing

FIG. 35

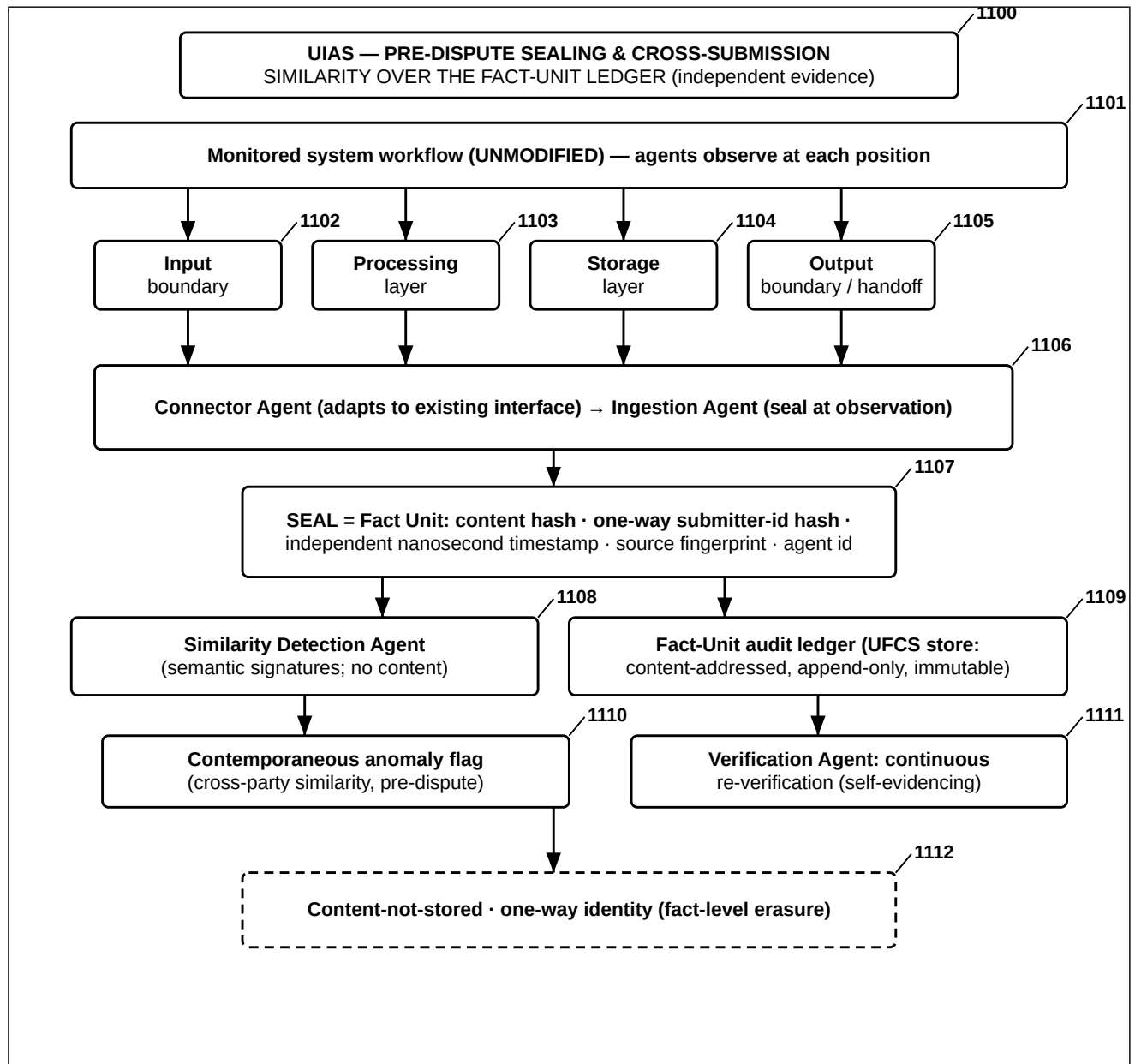


FIG. 35 — UIAS — Pre-Dispute Sealing and Cross-Submission Similarity over the Fact-Unit Ledger

FIG. 36

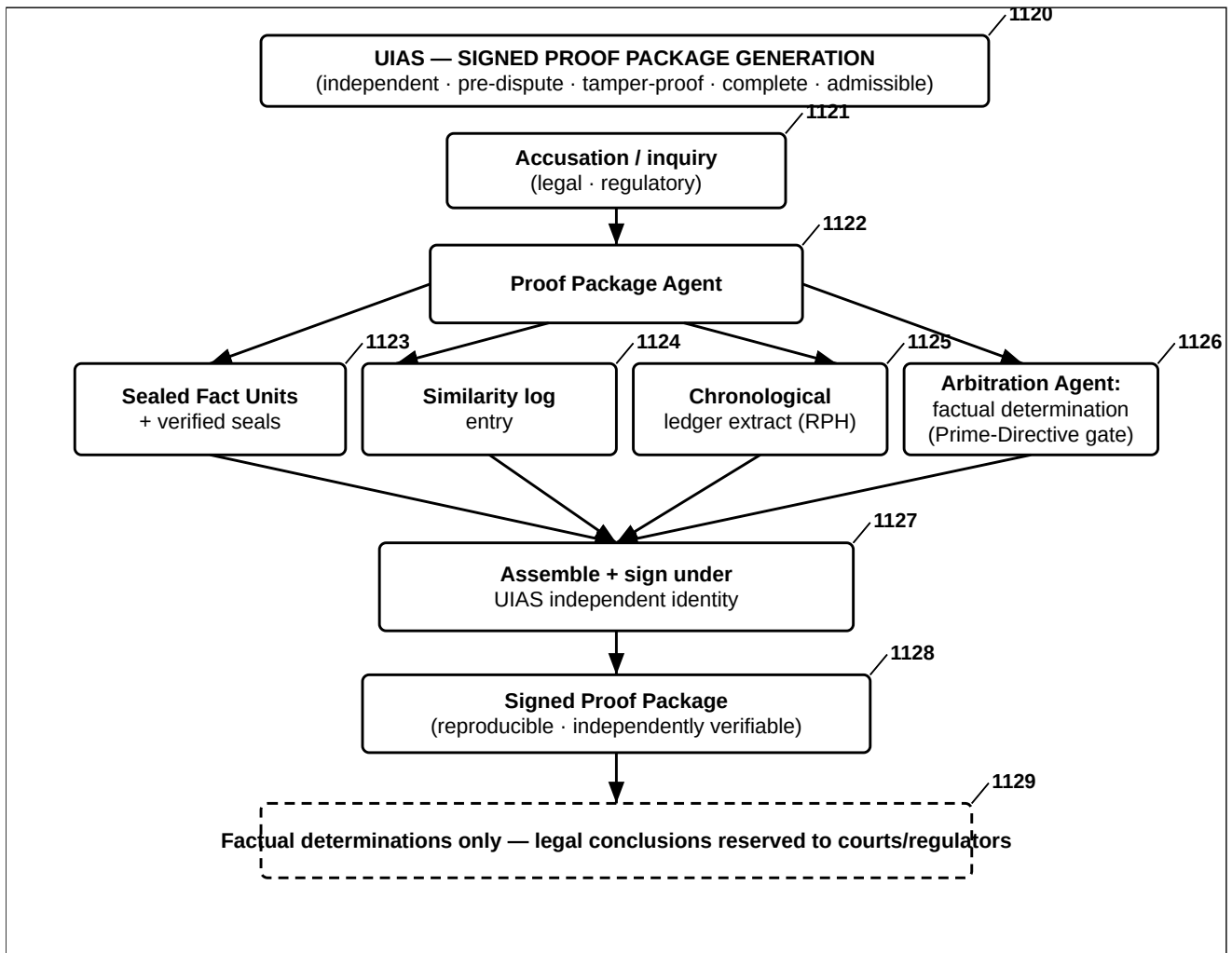


FIG. 36 — UIAS — Signed Proof Package Generation

FIG. 37

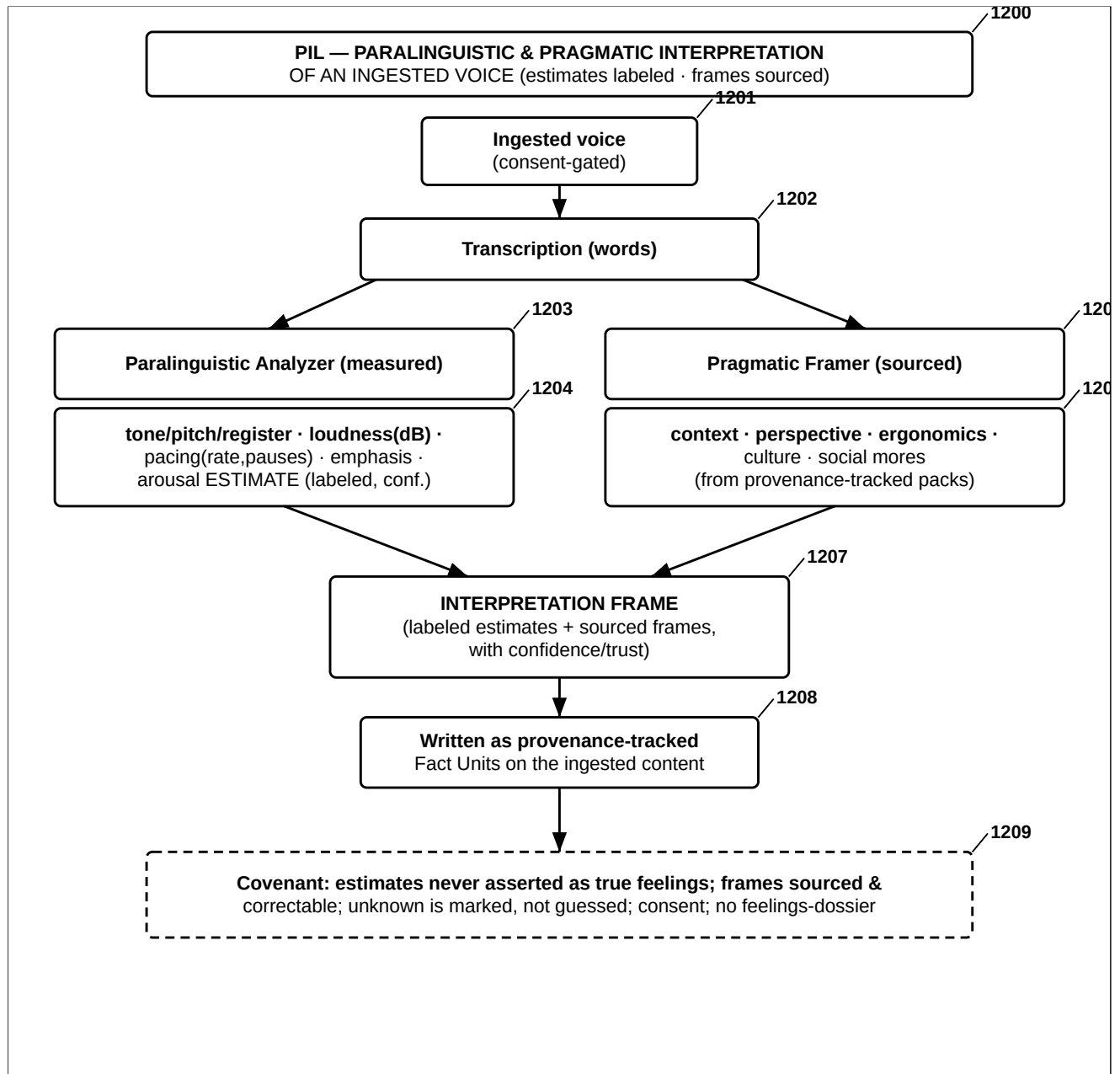


FIG. 37 — PIL — Paralinguistic and Pragmatic Interpretation of an Ingested Voice

FIG. 38

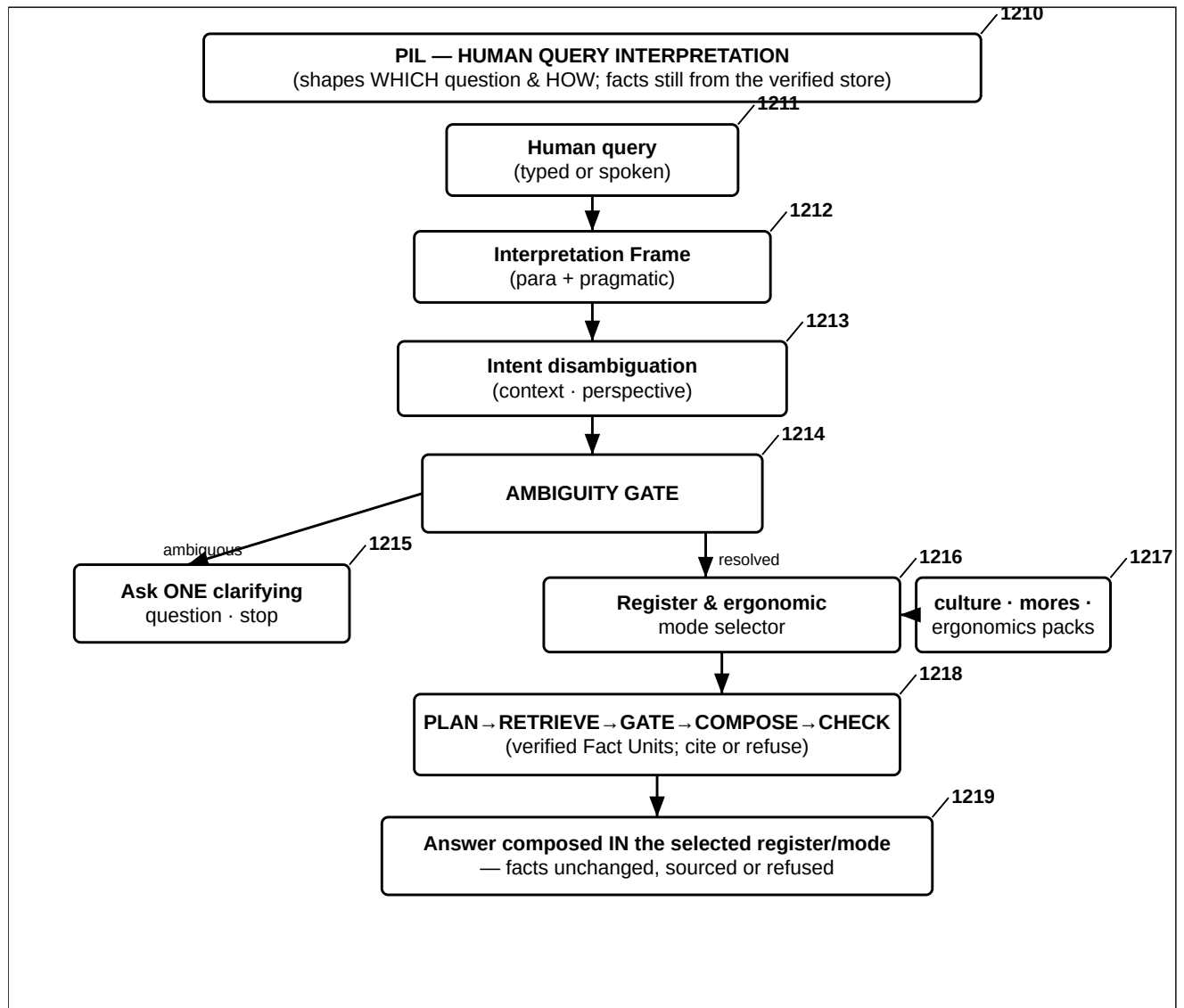


FIG. 38 — PIL — Human Query Interpretation and Covenant-Bound Answering

FIG. 39

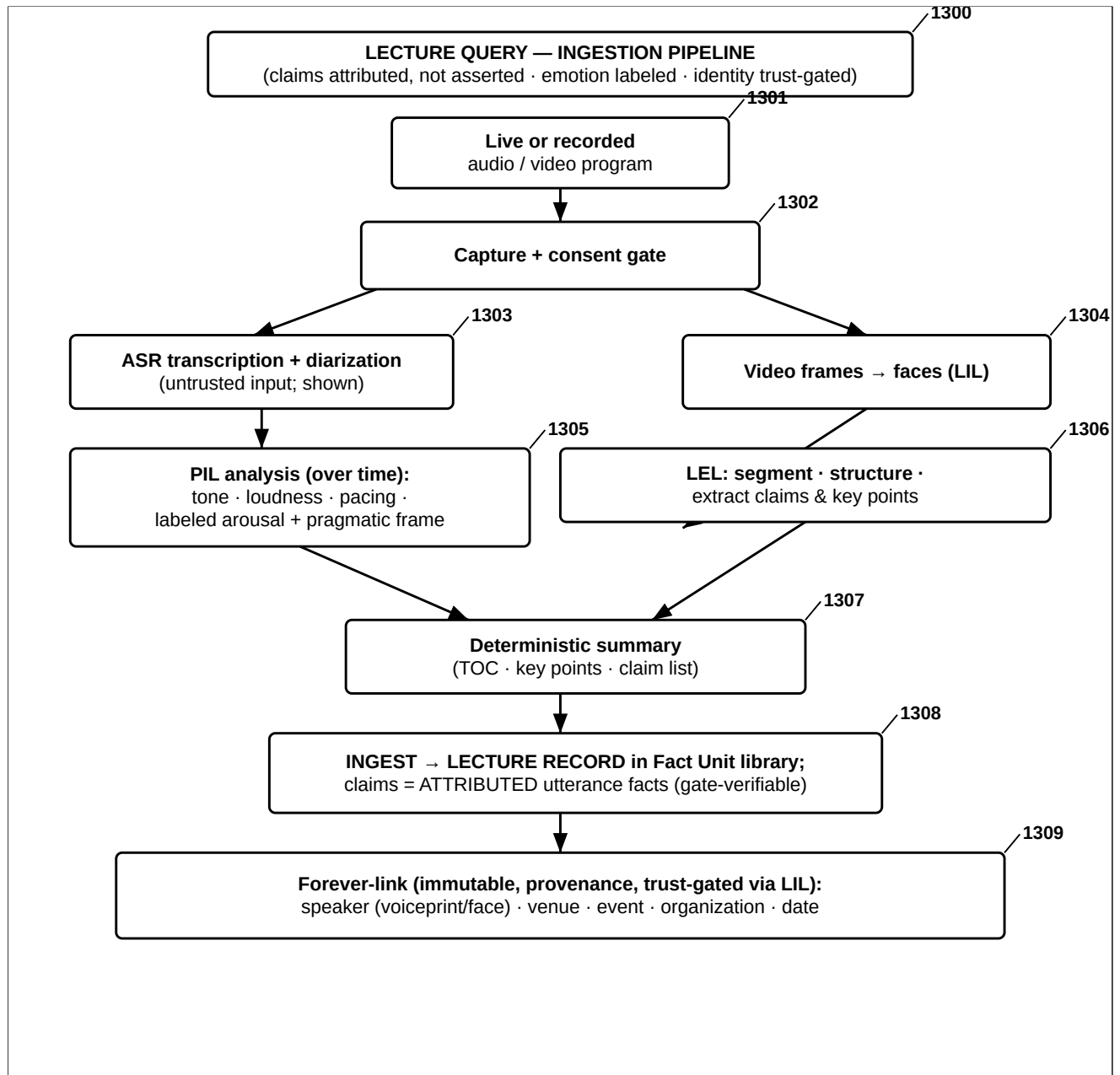


FIG. 39 — Lecture Query — Ingestion Pipeline

FIG. 40

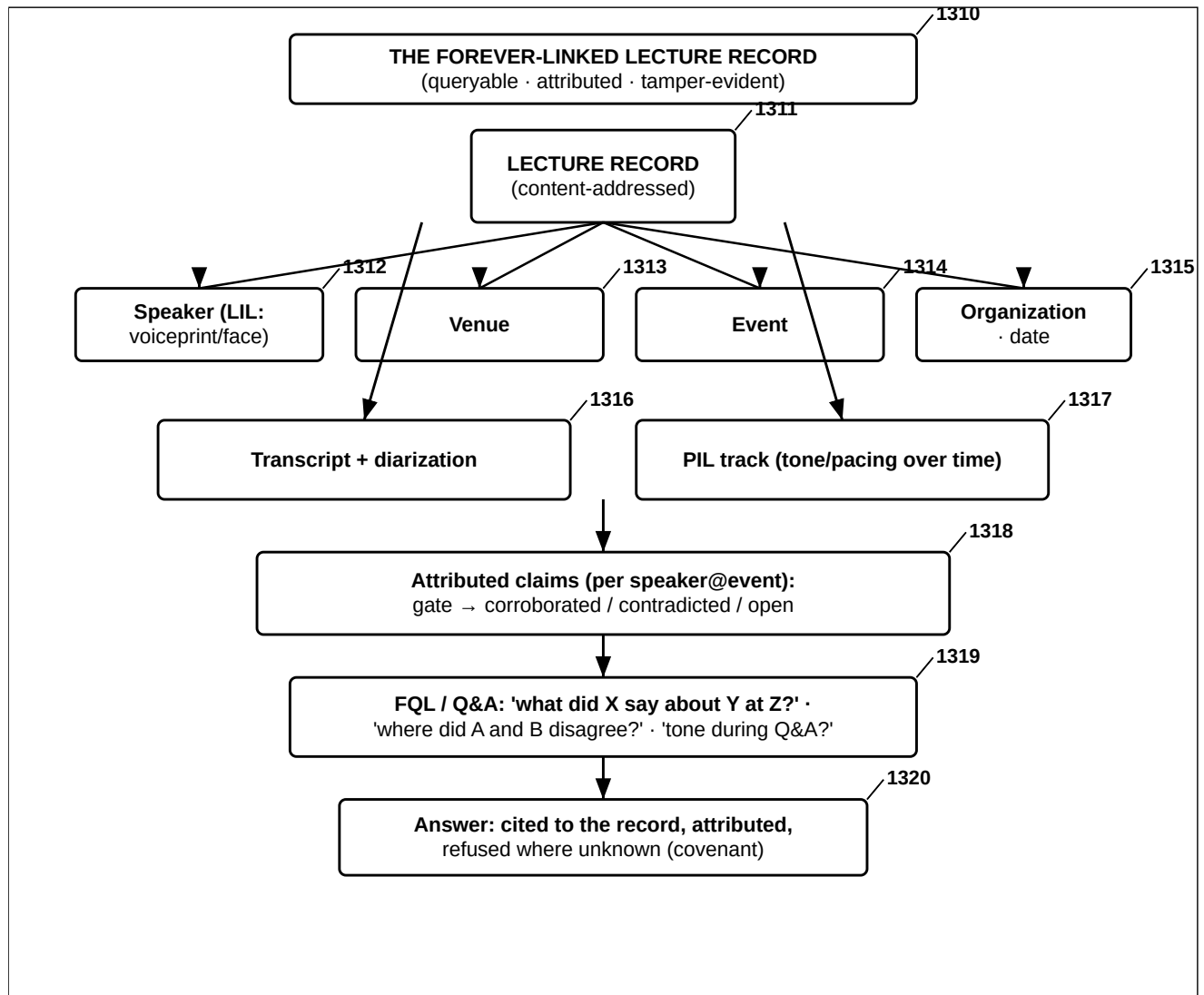


FIG. 40 — Lecture Query — The Forever-Linked Lecture Record and Its Querying

DRAFT — NOT FILED — FOR REVIEW BY QUALIFIED PATENT COUNSEL PRIOR TO FILING. Companion drawings volume to the consolidated QueryBook Provisional Patent Application; figures and numerals are drawn from, and identical to, that application. Finalize to 37 CFR 1.84 before filing.